

Co-design of Silicon Microring Modulator beyond 200 Gb/s per Lane: Device Physics, Operating Point, and Compact Models for Scale-Up and Scale-Out Optical I/O

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As artificial intelligence clusters scale to millions of accelerators, computing throughput is increasingly limited by the bandwidth density and energy efficiency of the optical interconnect fabric. Depletion-mode silicon microring modulators provide an essential path forward, combining femtojoule-per-bit junction switching energy with dense wavelength-division multiplexing. At these baud rates, device physics interacts directly with driver electronics, receiver noise, and link equalization. This review establishes a device-to-link pathway from 200 Gb/s PAM4 transmitters to the emerging 400G-per-lane frontier, showing that the optical operating point is an end-to-end link trade-off rather than an isolated modulator property. To enable unified electronic–photonic co-design, we evaluate how optical self-heating, free-carrier dispersion, and ambient temperature drift shift the resonance during modulation, setting quantitative requirements on cold-resonance placement and thermal tuning reserves. We deliver a calibrated physical and Verilog-A modeling framework that captures these coupled dynamics in circuit simulation, equipping system architects and transceiver designers to optimize 200G optical engines with predictable margins while establishing a foundation for 400G-per-lane architectures.

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I. INTRODUCTION

The scaling of artificial intelligence systems is constrained by the bandwidth and energy cost of moving data among accelerators. Training a frontier model requires tens of thousands of accelerators to operate as a single machine, but their computational throughput has grown faster than the bandwidth available for communication. Consequently, accelerator utilization can be limited by data movement among processors, memory and racks more than by computation^{1,2}.

Meeting this communication demand with electrical interconnects becomes more difficult as per-lane rates rise from 100 toward 200 Gb/s and beyond. Signals propagating through copper package and board channels undergo frequency-dependent attenuation, crosstalk and reflections, requiring equalization to compensate for channel loss and intersymbol interference (ISI). Increasing the number of lanes cannot remove these constraints indefinitely, because switch front-panel area and accelerator input/output (I/O) perimeter also limit the available connections.

Optical interconnects offer a route beyond these electrical reach constraints, with low fiber propagation loss over intra- and inter-rack distances. Their integration into computing systems has been supported by two decades of silicon-photonics development, through which modulators, photodetectors and passive circuits have reached volume manufacturing in CMOS-compatible processes^{3–5}. To shorten the remaining electrical path, optical engines are moving from faceplate pluggables toward near-package and co-packaged optics, millimeters from the switch or accelerator die^{6–9}. Coupling and component losses nevertheless remain part of the optical power budget.

Replacing the transmission medium does not, however, eliminate the bandwidth and energy constraints within the transceiver. At 200 Gb/s per lane, the modulator, photodetector and associated circuits can limit the link response, so signal processing remains necessary to achieve the target error rate. Multi-tap equalization and forward-error correction (FEC) in the DSP consume electrical power and add latency. The energy advantage of optical transmission therefore depends on how the device response and electronic signal processing are designed together, not on propagation loss alone.

Over the last decade, the Large-Scale Integrated Photonics group at Hewlett Packard Labs has developed a family of silicon photonics components for such links^{10–15}. On the optical source side, heterogeneous hybrid-bonded III–V-on-silicon microring lasers and quantum-dot frequency comb lasers provide energy-efficient, multi-wavelength optical carriers across dense channel grids^{11,14,15}. High-speed depletion-mode silicon microring modulators use resonant enhance-

ment to obtain low-voltage modulation, supporting 200 Gb/s PAM4 operation with 49 GHz of electro-optic bandwidth and femtojoule-scale junction switching energy per bit¹². Their performance nevertheless depends on the interplay between junction efficiency, cavity lifetime and laser detuning. This review focuses on those device-level trade-offs and the operating conditions needed to obtain useful modulation. On the receiver side, silicon/silicon-germanium APDs provide internal avalanche multiplication gain, which can improve sensitivity and reduce the required optical power^{16,17}; the avalanche receiver and link-level co-design are treated in a companion review.

This review is organized by first placing the transmitter within the landscape of AI interconnects, where scale-up and scale-out fabrics impose different requirements (Sec. II). Junction and cavity physics are then used to relate coupling and laser detuning to modulation above 200 Gb/s per lane, distinguishing the operating points with optimal transmission power (Sec. III). Because optical absorption also shifts the resonance, these operating points must be considered together with nonlinear optical effects and heater tuning (Secs. III C and IV). The electrical, optical and thermal equations provide a basis for translation into Verilog-A for system level simulation. Numerically calculated PAM4 eyes show how device response and operating point affect the transmitted waveform (Sec. V).

II. THE AI OPTICAL-INTERCONNECT LANDSCAPE

The interconnect architecture of artificial intelligence computing clusters is driven by fundamental memory and bandwidth scaling limits. Frontier neural network models, spanning hundreds of billions to trillions of parameters, exceed the memory capacity and compute throughput of any single accelerator chip. Distributing training across thousands of processors splits communication into distinct physical domains categorized by reach and loss constraints (Fig. 1). Scale-up links span millimeters on-package to several meters within a server chassis or rack, where tensor and pipeline parallelism demand continuous, low-latency all-reduce exchanges constrained by shoreline density and energy dissipation. Scale-out fabrics extend from 1 m to 2 km (and up to 10 km in campus networks), connecting accelerators and switch tiers across racks and rows under strict optical loss budgets. Beyond the cluster, scale-across networks span tens to thousands of kilometers between distant datacenters using coherent optical transport. Whether an optical link serves scale-up or scale-out determines the optimum modulator junction design, drive swing, and optical operating point.

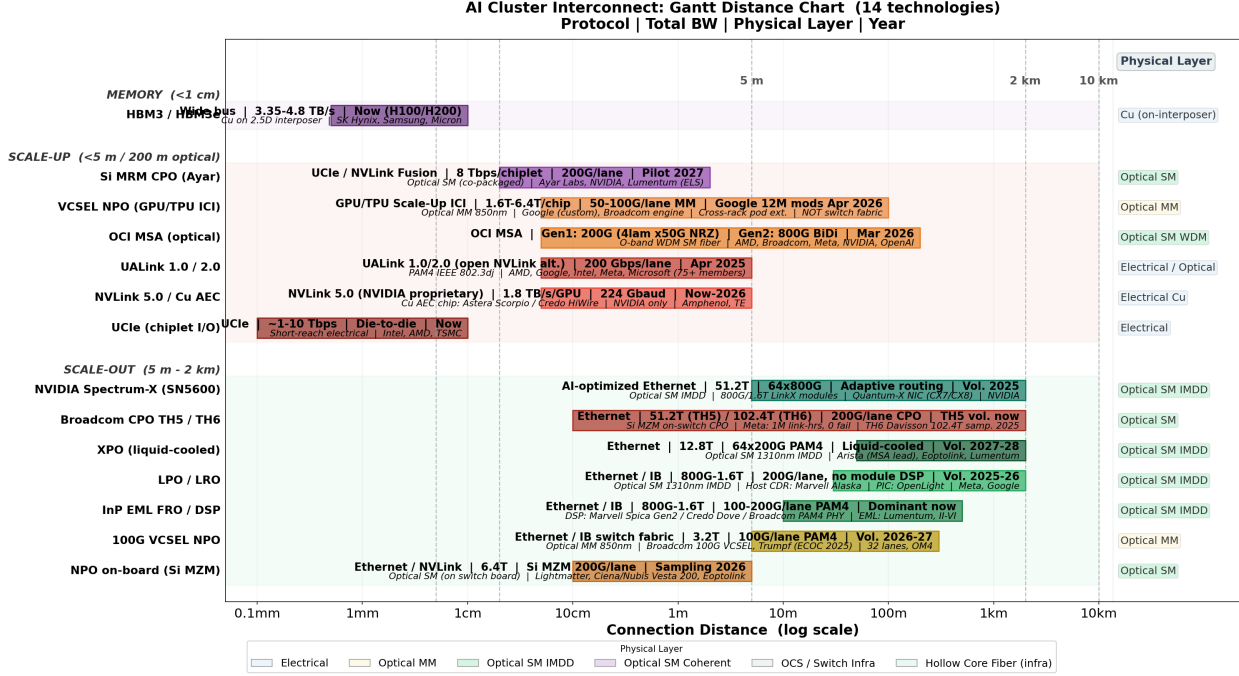


FIG. 1. AI interconnect technologies organized by connection distance and physical layer.

A. Scale-up fabrics and shoreline constraints

Scale-up fabrics link accelerators into a single, tightly synchronized logical computer, typically within a server chassis or rack. Current accelerators exchange tensor parameters over proprietary fabrics such as NVLink or open standards such as UALink. Emerging die-to-die interfaces such as UCle and memory protocols such as CXL over PCIe address complementary on-package and host-attached communication. At a lane rate of 200 Gb/s per unidirectional channel, an aggregate bidirectional bandwidth of 1.8 TB/s requires 72 physical channels per package, or 36 in each direction for a symmetric interface; a 3.6 TB/s target requires 144 channels in total. During distributed neural network training, collective communication routines such as All-Reduce, All-Gather, and Reduce-Scatter require all participating processors to exchange intermediate gradients before subsequent forward and backward computation can proceed. Because every processor must complete its exchange before the collective barrier clears, tail latency and link-to-link skew directly stall compute pipelines, making high-latency, power-intensive digital signal processors (DSPs) unacceptable in scale-up networks.

Electrical channels traversing packages and printed circuit boards cannot support these data rates across meter-scale distances. At symbol rates exceeding 100 GBaud, copper traces suffer

severe dielectric attenuation, crosstalk, and impedance discontinuities, restricting high-speed electrical reach to tens of centimeters. Thermal dissipation and package physical geometry impose further restrictions. High-Bandwidth Memory (HBM) stacks occupy nearly the entire perimeter of the accelerator package, leaving millimeter-scale shoreline windows for external input/output (I/O). Escaping terabytes per second through this congested edge requires ultra-high bandwidth density in terabits per second per millimeter of package shoreline.

Depletion-mode silicon microring modulators provide an ideal physical match for scale-up fabrics. Their compact footprint and small junction capacitance enable multi-channel dense wavelength-division multiplexing (DWDM) across 8 to 16 wavelengths on a single bus waveguide and single fiber, without requiring bulky arrayed waveguide gratings or external optical multiplexers. Furthermore, the small junction area yields switching energies below 100 fJ/bit, enabling direct driving from low-voltage CMOS electronics without power-hungry retimers. However thermal stability remains a challenge for microrings. Compute accelerators dissipate hundreds of watts, generating severe localized thermal flux and dynamic junction temperature excursions. Under the OIF 3.2 Tb/s co-packaged optics implementation agreement¹⁸, optical engines without integrated lasers operate across a case temperature range of 15 to 85 °C. With silicon exhibiting an 80 pm/K thermo-optic resonance shift¹⁹, uncompensated temperature drift detunes the optical cavity from the laser wavelength, degrading modulation depth and closing the optical eye. Consequently, scale-up transmitters require integrated micro-heaters and active closed-loop thermal stabilization²⁰.

B. Scale-out fabrics and optical link budgets

Scale-out fabrics connect switch application-specific integrated circuits (ASICs) and server nodes across racks, rows, and datacenter clusters, spanning reaches from 1 m to 2 km, and up to 10 km for campus-wide links. Pluggable optical transceivers, near-package optics (NPO), and co-packaged optics (CPO) differ in the physical separation between the optical engine and the switch ASIC. Moving the optics closer to the switch silicon shortens the host electrical channel and reduces electrical energy dissipation, while pluggable transceivers offer modularity and straightforward field servicing^{6–8}.

In scale-out interfaces, the high-speed electrical signal travels across the host printed circuit board (PCB) to the optical transceiver or co-packaged engine, where it converts into an optical

signal. The optical waveform then propagates through optical connectors and single-mode fibers across racks and switch tiers to reach destination servers. Unlike scale-up links where package shoreline density dominates, scale-out links are governed by the optical channel loss budget. Optical connector insertions and fiber attenuation across 100 m to 2 km paths introduce 4 to 6 dB of optical channel loss in standard 200GBASE-DR4 and FR4 specifications. To close this link budget under interoperability standards such as IEEE P802.3dj for 200 Gb/s per lane²¹ and OIF CEI-224G²², the optical transmitter must deliver high launch optical modulation amplitude (OMA) while satisfying strict transmitter and dispersion eye closure quaternary (TDECQ) limits.

In this loss-dominated regime, competing optical modulator technologies present distinct performance trade-offs. Indium phosphide (InP) electro-absorption modulators (EAMs) have demonstrated 155 GBaud PAM4 at 310 Gb/s and 400 Gb/s using PAM6²³, and heterogeneous InP modulators on silicon have achieved 170 GBaud PAM4 over 500 m of optical fiber²⁴. Thin-film lithium niobate (TFLN) modulators deliver electro-optic bandwidths beyond 100 GHz with low drive voltages and high linearity^{25,26}. Plasmonic modulators provide wide analog bandwidth and femtofarad capacitance at the cost of high insertion loss^{27,28}. Silicon Mach–Zehnder and microring modulators must likewise meet the bandwidth and drive-voltage requirements for operation at these rates.

When deployed in scale-out fabrics, the microring transmitter can no longer be operated solely for low drive swing. Instead, the operating point must shift toward the maximum dynamic OMA regime, where optical cavity peaking deliberately compensates for high-frequency electrical roll-off. Although this operating point introduces transient overshoot and higher raw eye distortion, receive-side feed-forward equalization (FFE) or digital signal processing recovers the eye margin²⁹.

Beyond the cluster switch fabric, scale-across and long-haul links connect distributed datacenters across metropolitan and continental distances. These networks use coherent optical transceivers encoding dual-polarization amplitude and phase under OIF 800ZR and 1600ZR implementation agreements^{30,31}. Because coherent systems rely on external local oscillators, complex IQ modulators, and full-retiming DSPs, their receiver architecture differs fundamentally from the direct-detection intensity-modulated links addressed here. The divergent constraints across these interconnect regimes motivate the integrated transceiver architecture and analytical modeling framework developed in Section III.

III. ANALYTICAL MODELING OF SILICON MICRORING MODULATORS

The divergent constraints of scale-up and scale-out demonstrate that an optical modulator cannot be evaluated or optimized as an isolated device. In scale-up fabrics, high shoreline bandwidth density, low driver dissipation, and self-restoring thermal stability govern the transceiver, favoring compact, low-capacitance resonators. In scale-out fabrics, launch optical modulation amplitude and receiver sensitivity dominate, requiring transmitters to close multi-decibel optical loss budgets across hundreds of meters to kilometers of fiber. Translating these conflicting network demands into quantitative device specifications requires analyzing the modulator, channel, and receiver as an integrated link.

Figure 2 shows an end-to-end optical link combining a depletion-mode silicon microring modulator with an avalanche photodiode (APD) receiver. The depletion-mode microring modulator converts a voltage-induced resonance shift into an intensity change when biased on a resonance flank. Resonant enhancement enables high-speed modulation within a sub-15 μm cavity, reducing active junction length and lowering capacitive switching energy to femtojoule-per-bit levels. This switching-energy advantage must be evaluated against the optical modulation amplitude (OMA) and bandwidth delivered to the channel, accounting for junction parasitics, driver swing, and optical insertion loss.

On the receive side, the APD provides internal avalanche multiplication gain M before the transimpedance amplifier, suppressing the input-referred thermal noise of the electronic front end. When electronic amplifier noise dominates, avalanche multiplication improves receiver sensitivity, directly relaxing the transmitter launch OMA required to achieve the target bit-error rate across the link. Conversely, in low-loss scale-up links where optical power is preserved, optical engines can prioritize low driver swing and minimal equalization latency. The analytical framework developed in the following sections resolves these interrelated device, cavity, and circuit dynamics, beginning with junction design and active carrier-mode overlap.

A. High-speed silicon microring modulators

Si depletion-mode microring modulators have demonstrated 128 Gb/s PAM4 with integrated thermo-optic tuning³² and 240 Gb/s on lateral³³ and L-shaped junction platforms³⁴. More recent work reports 400 Gb/s PAM6 eyes with DSP and acquisition averaging³⁵.

Illustrative microring and APD link

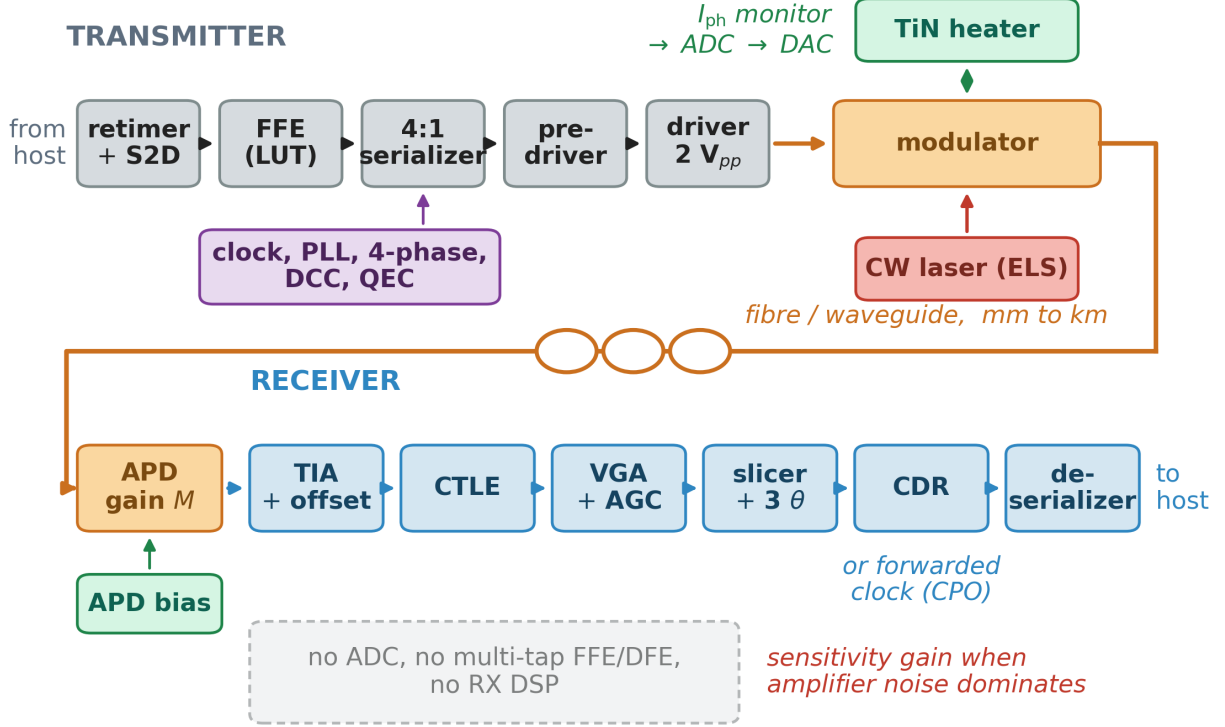


FIG. 2. Illustrative MRM–APD link with analog receiver equalization and slicing. Thermal-loop converters are outside the data path.

Table I compares four high-speed Si microring modulators, including our 5×200 Gb/s PAM4 Z-shaped microring modulator¹², an eight-channel PAM4 transmitter using 256G lateral rings³⁶, a 16×128 Gb/s device³⁷ and a lateral two-segment device³⁸. The devices differ in junction overlap, operating point, and the balance between per-lane rate and channel count.

The junction cross-section sets the overlap Γ between the optical mode and the depletion region. At a common depletion width of 40 nm, the calculated overlaps are $\sim 14.7\%$ for a lateral junction, $\sim 25.5\%$ for a vertical junction, $\sim 25.9\%$ for an L-shaped junction and $\sim 28.7\%$ for a Z-shaped junction [Fig. 3(a)–(d)]¹². A larger overlap improves the modulation efficiency at a given junction capacitance. Compared with a lateral junction of $V_{\pi}L \sim 1.00 \text{ V}\cdot\text{cm}$ ³⁹, the Z-shaped junction reaches $\sim 0.67 \text{ V}\cdot\text{cm}$, corresponding to higher phase-modulation efficiency, and doubles the data rate at $\sim 36\%$ lower junction-switching energy per bit¹². The three devices in Table I with reported $V_{\pi}L$ values span a narrow range of 0.63–0.69 V·cm despite their different junction geome-

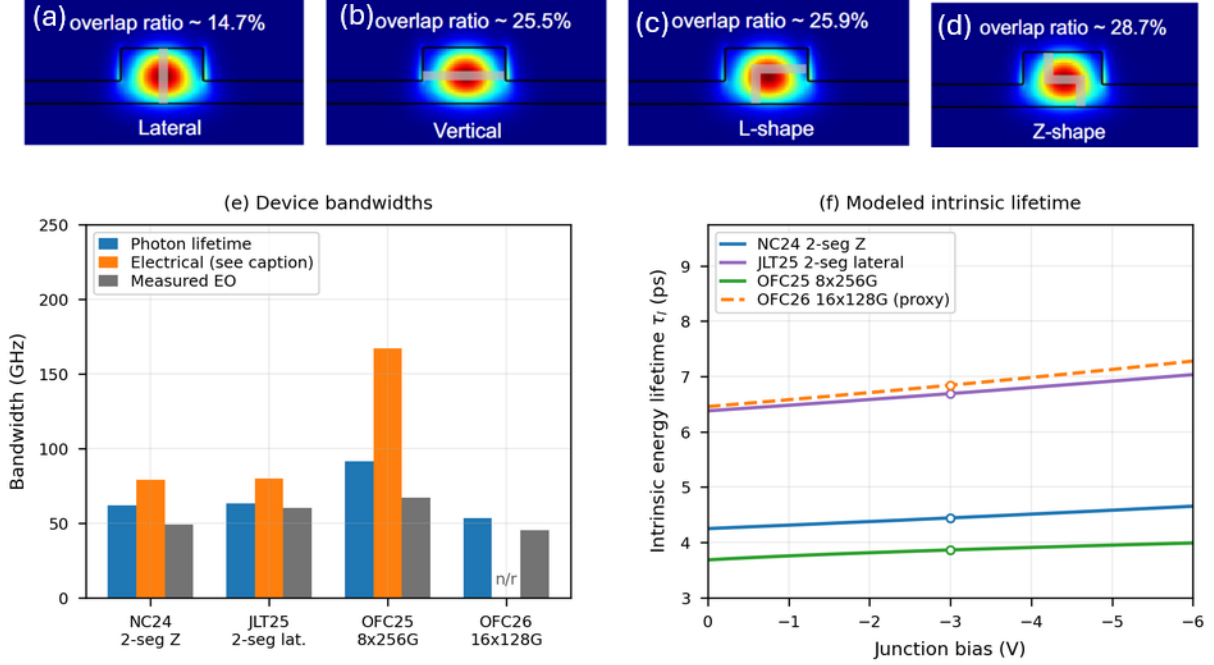


FIG. 3. Junction comparison. **(a)–(d)** Simulated modes and depletion overlaps. **(e)** Photon-lifetime, electrical and measured EO bandwidths from Table I, taking the shorter-segment electrical value for each two-segment device, where n/r denotes not reported. **(f)** Modeled intrinsic energy lifetime versus bias. Circles mark assumed -3 V references; the dashed 128 Gb/s curve uses a doping proxy.

tries. Reduced optical overlap can be compensated by higher junction doping, at the expense of increased junction capacitance per unit length. Increasing reverse bias reduces carrier absorption and lengthens the intrinsic energy lifetime τ_i [Fig. 3(f)].

Junction doping sets the depletion width and hence both the tuning efficiency and C_j . Higher doping narrows the depletion region and increases the capacitance, while the usable reverse-bias swing remains constrained by electric field, leakage and breakdown. The doping profile must therefore balance tuning efficiency, capacitance and optical loss at the intended drive conditions. The Z-shaped ring’s upper junction region has p-type and n-type concentrations of $3 \times 10^{18} \text{ cm}^{-3}$ each¹². The 256G lateral ring in the eight-channel transmitter uses an n-type concentration of 10^{19} cm^{-3} and a p-type concentration of $5.3 \times 10^{18} \text{ cm}^{-3}$ ³⁶. The reconstructed series resistance is $\sim 16 \Omega$ for the 256G lateral ring in the eight-channel transmitter, compared with a fitted $\sim 386 \Omega$ for the shorter segment of the lateral two-segment ring³⁸, as specified in Table I.

Table I groups the devices by junction segmentation. Single-segment microring modulators

drive a continuous junction around the ring. Direct PAM4 modulation requires a four-level electrical waveform, commonly generated by a DAC and driver, with increasing power and equalization demands at higher rates³⁶. Two-segment designs instead generate four optical levels from two NRZ drives for the least significant bit (LSB) and most significant bit (MSB), using nominal modulation weights of one and two^{12,39}. Segmentation also reduces the load on each driver. The Z-shaped segments have capacitances of ~ 6.6 and ~ 13.2 fF, against ~ 19.8 fF for an equivalent one-segment junction. Including the source and fitted parasitic network, the corresponding electrical bandwidths increase from ~ 54 GHz for the one-segment case to ~ 79 and ~ 65 GHz for the shorter and longer segments, respectively¹². The bandwidth improvement follows from the smaller segment loads within the fitted network.

The Z-shaped ring combines enhanced junction overlap with two small capacitive loads, demonstrating 100 GBaud PAM4 with two 1.6 V binary drives. The 256G lateral ring reaches 128 GBaud with a 1.8 V PAM4 drive (Table I). These architectures connect junction efficiency, cavity coupling and driver loading to the requirements for 200G-class operation.

B. Depletion-mode silicon microring modulators

1. Coupled-mode theory

A microring is resonant when its round-trip phase is an integer number of cycles^{40,41}. Interference between the bus and circulating fields converts resonance tuning into intensity modulation. The coupled-mode description relates resonator–waveguide coupling⁴² to RF-to-optical conversion⁴³, with the cavity response⁴⁴

$$\frac{da(t)}{dt} = \left[j\omega_r - \frac{1}{2\tau_p} \right] a(t) + \frac{1}{\sqrt{\tau_e}} E_{\text{in}}(t), \quad E_{\text{out}}(t) = E_{\text{in}}(t) - \frac{1}{\sqrt{\tau_e}} a(t), \quad (1)$$

where $a(t)$ is the cavity amplitude and $|a(t)|^2$ the stored energy. The bus fields $E_{\text{in}}(t)$ and $E_{\text{out}}(t)$ are normalized so that $|E(t)|^2$ is optical power. The resonance angular frequency is $\omega_r = 2\pi c/\lambda_{\text{res}}$, where λ_{res} is the resonance wavelength. The energy lifetimes τ_e and τ_l describe coupling and intrinsic loss, with combined photon lifetime τ_p . After removing the input at $t = 0$, fixed ω_r and τ_p give

$$a(t) = a(0)e^{j\omega_r t} e^{-t/(2\tau_p)}, \quad |a(t)|^2 = |a(0)|^2 e^{-t/\tau_p}. \quad (2)$$

TABLE I. Microring device comparison. Paired values denote shorter/longer segments; swings are per segment and bias points denote laser detuning.

Parameter	Two-segment		Single-segment lateral	
	Z-shaped ¹²	Lateral ³⁸	8 × 256 Gb/s ³⁶	16 × 128 Gb/s ³⁷
Ring radius (μm)	12	6	6	6
Loaded Q_L	3700	3700	2500	4300
DC extinction ratio (dB)	16	—	>20	>18
Coupling condition	—	over (12% target)	near-critical	near-critical
Tuning efficiency (pm/V)	27	7 / 15	33	27
$V_\pi L$ (V·cm)	0.67	0.63	0.69	—
C_j (fF)	6.6 / 13.2	6.0 / 10.0	~20*	—
C_{pad} (fF)	31.6 / 33.1	24.2 / 24.6	~36*	—
R_s (Ω)	68.1 / 35.4	385.9 / 246.7	~16*	—
Electrical bandwidth (GHz)	79 / 65	80 / 75	167*	—
f_{ph} (GHz)	62	63	92	53
f_{EO} (GHz)	49 / 48	60	67	45
Bias point	max OMA	near max GBW	near max GBW	near max GBW
Drive swing (V_{pp})	1.6	~2	1.8	1.6
Symbol rate (GBaud)	100	64	128	64
TDECQ (dB)	—	—	2.6–2.9	2.7–2.9

*Calculated from this paper.

The stored energy falls to $1/e$ of its initial value after τ_p , whereas the field magnitude requires $2\tau_p$. Longer storage narrows the resonance, relating these lifetimes to the quality factors,

$$Q_L \equiv \frac{\lambda_{\text{res}}}{\Delta\lambda_{\text{FWHM}}} = \omega_r \tau_p = \frac{2\pi c \tau_p}{\lambda_{\text{res}}}, \quad Q_i = \omega_r \tau_l, \quad Q_c = \omega_r \tau_e \quad (3)$$

where Q_L , Q_i and Q_c are the loaded, intrinsic and coupling quality factors, and $\Delta\lambda_{\text{FWHM}}$ is the

full width at half the notch depth in linear power. The two energy-decay rates add, giving

$$f_{\text{ph}} = \frac{c}{\lambda_{\text{res}} Q_L} = \frac{1}{2\pi\tau_p}, \quad \frac{1}{\tau_p} = \frac{1}{\tau_e} + \frac{1}{\tau_l}. \quad (4)$$

where f_{ph} is the linewidth-derived photon-lifetime frequency used in the approximate bandwidth budget. A 67 GHz EO target requires $\tau_p < 2.4$ ps within the bandwidth budget of Eq. (17), with additional allowance for junction charging. At 1310 nm, the Z-shaped ring's loaded quality factor $Q_L = 3700$ gives 2.6 ps and 62 GHz, compared with 1.7 ps and 92 GHz for the 256G lateral ring at $Q_L = 2500$. Their measured EO bandwidths are 49/48 GHz for the Z-shaped segments and 67 GHz for the 256G lateral ring (Table I). Enhanced junction overlap provides efficient tuning, while stronger bus coupling shortens the photon lifetime. The intrinsic lifetime τ_l is determined by free-carrier absorption, scattering and bending radiation. Reverse-bias depletion reduces absorption and lengthens this lifetime, while self-heating changes both loss and resonance position (Sec. III C). The external lifetime τ_e depends on the energy transferred to the bus per round trip.

At resonance, the extent of destructive interference is determined by the balance between external coupling and intrinsic loss,

$$\rho \equiv \frac{\tau_l}{\tau_e} = \frac{Q_i}{Q_c}, \quad D = 1 - \left(\frac{1 - \rho}{1 + \rho} \right)^2, \quad \text{ER}_{\text{notch}} = -10 \log_{10}(1 - D). \quad (5)$$

where ρ is the ratio of external to intrinsic decay rates, D the fractional on-resonance power extinction, and ER_{notch} the static notch ER. Critical coupling gives complete cancellation in this ideal model. The Z-shaped ring has a 16 dB notch ER, while the 256G lateral ring exceeds 20 dB, requiring $D \geq 99\%$. On the slightly overcoupled branch where the coupling Q_c is less than the intrinsic Q_i , stronger coupling broadens the resonance at the expense of notch depth. At $\rho = 1.2$ and $Q_L = 2500$, the intrinsic and coupling quality factors Q_i and Q_c are approximately 5500 and 4583, respectively. The coupler power transfer is related to the external lifetime by

$$\kappa^2 \simeq \frac{n_g L_{\text{rt}}}{c \tau_e}, \quad \kappa^2 + |t_c|^2 = 1, \quad (6)$$

where κ^2 is the coupler power-transfer fraction, n_g the group index and L_{rt} the round-trip length. The lifetime approximation assumes weak coupling, and power conservation assumes a lossless coupler with field self-coupling coefficient t_c .

The photon lifetime sets the wavelength linewidth,

$$\Delta\lambda_{\text{FWHM}} \simeq \frac{\lambda_{\text{res}}^2}{2\pi c \tau_p}. \quad (7)$$

The round-trip group delay, independent of the decay time, sets the spacing between adjacent resonances,

$$\text{FSR} \simeq \frac{\lambda_{\text{res}}^2}{n_g L_{\text{rt}}}, \quad (8)$$

where FSR is the local wavelength spacing. For the 256G lateral ring, $n_g = 4.0$ and $L_{\text{rt}} = 38.7 \mu\text{m}$ give an 11.1 nm FSR at 1310 nm.

At a given group index, the target FSR fixes the round-trip length. A circular ring and a race-track with two equal straight sections then satisfy

$$R_{\text{ring}} \simeq \frac{\lambda_{\text{res}}^2}{2\pi n_g \text{FSR}} \quad (\text{circular ring}), \quad L_{\text{rt}} = 2\pi R_{\text{ring}} + 2L_s \quad (\text{racetrack}), \quad (9)$$

where R_{ring} is the bend radius and L_s is the length of each straight section. At fixed FSR and group index, a longer straight interaction requires smaller bends, linking coupling length to bending loss.

For a uniform, lossless, phase-matched straight coupler of length L_c , the even and odd supermodes give

$$\kappa^2 = \sin^2 \left[\frac{(\beta_+ - \beta_-)L_c}{2} \right], \quad (10)$$

where β_+ and β_- are the propagation constants of the even and odd supermodes. A smaller gap generally increases their splitting, and weak-transfer coupling grows quadratically with interaction length. Approach bends and phase mismatch must be included in the fabricated design.

For monochromatic input of wavelength λ and fixed cavity parameters, the cavity-only through-port field transmission is

$$t(\lambda) = \frac{E_{\text{out}}(t)}{E_{\text{in}}(t)} = 1 - \frac{1/\tau_e}{1/(2\tau_p) - j2\pi c(1/\lambda_{\text{res}} - 1/\lambda)}, \quad (11)$$

where λ is the laser wavelength and $t(\lambda)$ approaches unity far from resonance. The fields are defined at the bus ports in Eq. (1), with resonance-minus-laser detuning $\Delta\omega = \omega_r - \omega_{\text{in}} = 2\pi c(1/\lambda_{\text{res}} - 1/\lambda)$.

Including nonresonant loss between the chosen input and output reference planes gives the absolute power transmission $T(\lambda)$. With detuning normalized to the resonance half-linewidth,

$$x \equiv 2\Delta\omega \tau_p \simeq \frac{2(\lambda - \lambda_{\text{res}})}{\Delta\lambda_{\text{FWHM}}}, \quad T(\lambda) = T_{\text{off}}|t(\lambda)|^2 = T_{\text{off}} \left[1 - \frac{D}{1+x^2} \right], \quad (12)$$

where x is the dimensionless laser detuning and $T_{\text{off}} = |t_{\text{off-res}}|^2$ is the off-resonance power transmission. The off-resonance field ratio $t_{\text{off-res}}$ includes waveguide and taper losses and is treated as constant across the resonance. Positive unit detuning is one half-linewidth to the red, whereas negative detuning is blue.

The driven PAM4 extinction ratio (ER) compares the highest and lowest optical levels. The Z-shaped ring reports 3.6 dB outer ER at 100 GBaud under the conditions in Sec. V C 1. Operating closer to the notch can increase ER while reducing transmitted power and compressing lower-level spacing. The detuning must therefore provide sufficient outer OMA and four-level separation, with TDECQ and BER evaluated at the target baud rate.

2. Plasma dispersion and tuning efficiency

Plasma dispersion links changes in the free-carrier density to the refractive index and absorption of silicon. Reverse bias depletes carriers, increasing the effective index and reducing free-carrier absorption. At an electron density of $3 \times 10^{18} \text{ cm}^{-3}$ in n-type silicon and a hole density of $3 \times 10^{18} \text{ cm}^{-3}$ in p-type silicon, the O-band Soref–Bennett fit gives a hole-induced index change relative to undoped silicon about $1.9\times$ that of electrons^{45–47}. The stronger hole response favors overlap with the p-doped region. For a uniform segment, optical overlap and active ring fraction weight the material index change Δn , giving the resonance shift

$$\Delta n_{\text{eff}} = \frac{L_j}{L_{\text{rt}}} \Gamma \Delta n, \quad \Delta \lambda_{\text{res}} = \frac{\Delta n_{\text{eff}}}{n_g} \lambda_{\text{res}}, \quad (13)$$

where Γ is the transverse junction–mode overlap and L_j the active junction length. The index change Δn_{eff} is averaged over the round trip, with segment contributions added. Greater overlap increases the effective-index change and resonance shift for a given material response. Operation on the resonance flank converts this shift into intensity modulation^{48–50}.

Figure 4 compares calculated spectra for two coupling regimes with measured spectra from our Z-shaped 5×200 Gb/s transmitter and the eight-channel transmitter using 256G lateral rings. Panels (a,c) show the calculated response as the junction bias varies from 0 to −4 V. The two alternative coupling cases use $Q_L = 3700$ and a 16 dB notch at the assumed −3 V, 1310 nm reference, with both segments biased together and temperature and external coupling held fixed. Increasing reverse bias expands the depletion region, reducing free-carrier absorption within the optical mode. The intrinsic lifetime τ_i and quality factor Q_i therefore increase, while Q_c remains approximately fixed. In an undercoupled ring, $Q_i < Q_c$, the increase in Q_i brings the decay rates toward balance and deepens the notch until critical coupling is reached. In an overcoupled ring, $Q_i > Q_c$, the same increase in Q_i moves the decay rates farther from balance and reduces notch ER. The measured notch deepening in the Z-shaped ring in panel (b) is consistent with undercoupling

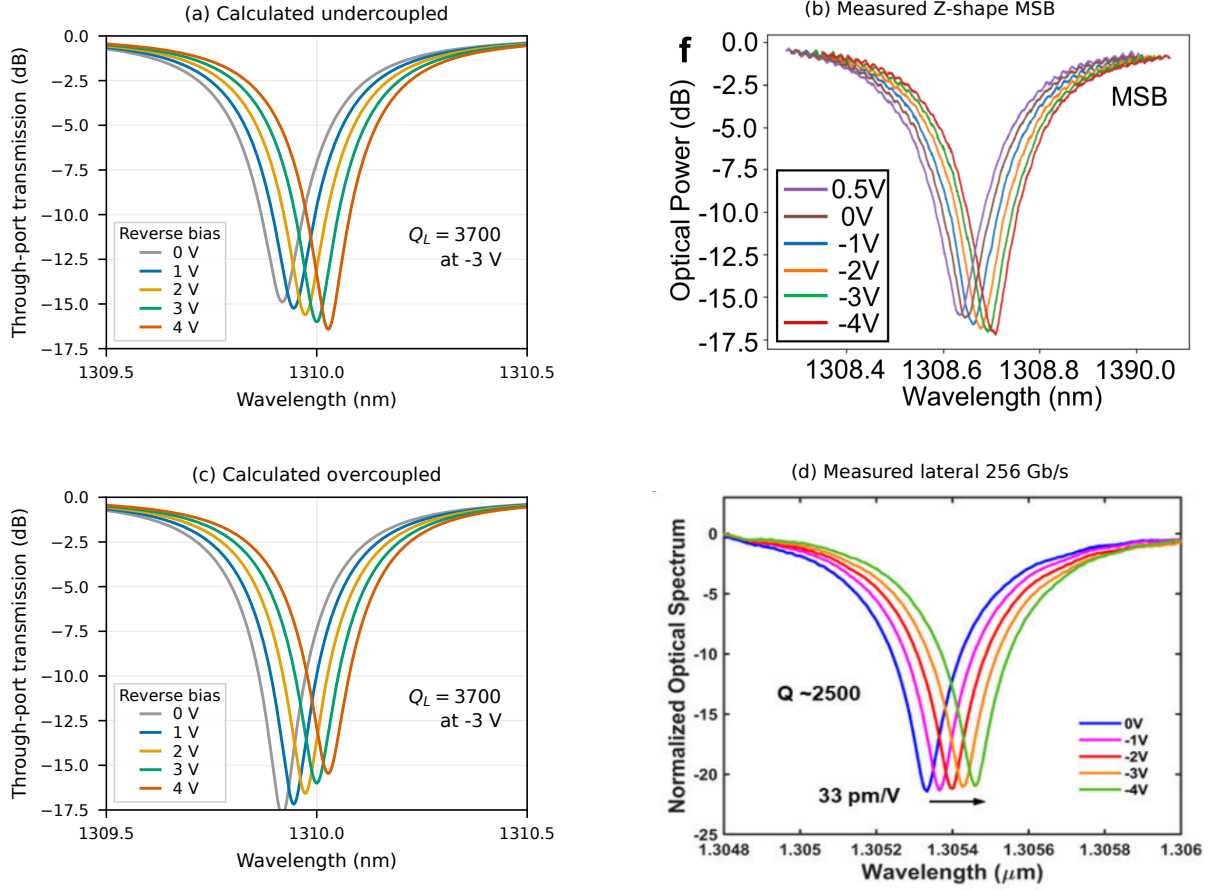


FIG. 4. Bias-dependent spectra. **(a,c)** Calculated undercoupled and overcoupled rings. **(b,d)** Measured Z-shaped ring and 256G lateral ring spectra, not model fits. Cropped/rescaled from Yuan *et al.*, Fig. 2(f)¹², CC BY 4.0, and Xue *et al.*, Fig. 1(c)³⁶, © 2025 Optica Publishing Group.

over the displayed reverse-bias range¹². The slight decrease in ER for the 256G lateral ring in panel (d) is consistent with weak overcoupling³⁶.

The tuning efficiency η is the resonance shift per volt of increased reverse bias. The voltage-length product $V_\pi L$ specifies the voltage and active length required for a π phase shift. Lower values indicate higher efficiency. The two metrics are related through the free spectral range,

$$\eta = \frac{\Delta\lambda_{\text{res}}}{\Delta V} = \frac{\text{FSR}}{2\pi} \frac{\Delta\phi_{\text{rt}}}{\Delta V} = \frac{\lambda_{\text{res}}}{n_g} \frac{L_j}{L_{\text{rt}}} \Gamma \frac{\partial n}{\partial V}, \quad (14)$$

$$V_\pi L = V_\pi L_j = \frac{\text{FSR} L_j}{2\eta} = \frac{\lambda_{\text{res}}^2 L_j}{2n_g L_{\text{rt}} \eta} = \frac{\lambda_{\text{res}}}{2\Gamma \partial n / \partial V}, \quad (15)$$

where ΔV is the increase in reverse-bias magnitude, $\Delta\phi_{\text{rt}}$ the round-trip phase shift and $\partial n / \partial V$

the material index response per volt. One FSR corresponds to 2π of round-trip phase, giving the factor of two in $V_\pi L$, a figure of merit depending on the junction cross-section. The Z-shaped ring's summed tuning efficiency of 27.3 pm/V at 1.6 V swing per segment gives a 44 pm extreme-state resonance shift. The 256G lateral ring gives a 59 pm shift at 33 pm/V and a 1.8 V swing. Both demonstrations used an arbitrary-waveform generator and linear amplifiers^{12,36}. Advanced-node core supplies of about 0.6 to 0.75 V require voltage-boosting output stages to deliver these swings from a co-packaged 5-, 3- or 2-nm driver. Junction overlap, doping and active length must provide the required resonance shift at the delivered voltage. Greater overlap improves tuning per unit capacitance, whereas increasing junction capacitance through doping, or increasing the active fraction at fixed ring length, can also raise tuning efficiency but increases electrical loading and adds doping-induced absorption to the optical-loss budget. Stacked output transistors distribute the swing across devices, while higher-voltage I/O transistors trade speed for voltage tolerance, requiring driver design to balance voltage sharing, bandwidth and power⁵¹. The DC bias and signal extrema, including overshoot, must satisfy junction leakage, breakdown and transistor terminal-voltage limits across process and temperature.

3. *Electro-optic bandwidth*

Junction doping controls depletion width and capacitance. Junction and access doping, together with contact geometry, determine series resistance R_s . For an abrupt junction with series access paths,

$$R_s = R_{n,\text{core}} + R_{p,\text{core}} + R_{n,\text{slab}} + R_{p,\text{slab}} + R_c + R_m, \quad C_j = \frac{\epsilon_{\text{Si}} h_j L_j}{W} = c_j L_j, \quad (16)$$

where the n- and p-side core and slab resistances, total contact resistance R_c and internal metal resistance R_m form the device series resistance. External driver and feed impedances are excluded from R_s . Here ϵ_{Si} is the silicon permittivity, h_j the effective junction-wall height, W the depletion width, and c_j the junction capacitance per unit length. At fixed reverse bias, heavier doping supplies more ionized dopant charge per unit volume, supporting the junction potential across a narrower depletion region and increasing capacitance. Increasing reverse bias expands the depletion region and lowers capacitance. Extending the active length increases both junction area and optical interaction length, increasing capacitance and phase shift together.

The electrical bandwidth $f_{\text{RC},\text{network}}$ is the 3 dB bandwidth of V_j/V_{src} relative to its low-frequency response. When one RC pole dominates, the lumped estimate $f_{\text{RC},\text{lump}} \simeq 1/(2\pi RC)$

applies, with effective loaded resistance R and capacitance C . The full network retains frequency-dependent driver, feed and shunt impedances (Sec. III B 6). Distributed or measured S-parameter models are needed when interconnect propagation and reflections are appreciable. With a $50\ \Omega$ source, the fitted networks give approximately 79 and 65 GHz for the Z-shaped LSB and MSB segments, respectively (Table I).

Combining loaded electrical and cavity responses gives the approximate inverse-square EO bandwidth budget,

$$\frac{1}{f_{\text{EO}}^2} \simeq \frac{1}{f_{\text{ph}}^2} + \frac{1}{f_{\text{RC},\text{network}}^2} = \left(\frac{\lambda_{\text{res}} Q_L}{c} \right)^2 + \frac{1}{f_{\text{RC},\text{network}}^2}, \quad (17)$$

where f_{EO} is an engineering estimate for approximately low-pass responses, using the loaded electrical bandwidth and photon-lifetime scale.

For the 67 GHz target of the 256G lateral ring, its 92 GHz photon-lifetime scale requires approximately 98 GHz loaded electrical bandwidth within this budget, equivalent to a 1.6 ps time constant including driver and parasitic loading.

4. *Efficiency–capacitance plane*

Figure 5 relates tuning efficiency to electrical loading. At fixed overlap, a larger capacitance displaces more charge per volt and reduces the voltage–length product ($V_\pi L$). With carrier density N and plasma-dispersion coefficient $\partial n / \partial N$, the local constant-capacitance approximation gives

$$V_\pi L \simeq \frac{K_0}{\Gamma c_j}, \quad K_0 = \frac{\lambda q A_{\text{mode}}}{2|\partial n / \partial N|} \quad (18)$$

where A_{mode} is the effective optical mode area, q the elementary charge, and K_0 collects the wavelength, area and carrier-response factors. At a 40 nm depletion width, overlaps of 14.7%, 25.5%, 25.9% and 28.7% for lateral, vertical, L-shaped and Z-shaped junctions¹² give a $1.9\times$ Z-shaped efficiency advantage over the lateral geometry at equal capacitance and fixed K_0 . The plotted normalization uses nominal one-third/two-thirds round-trip lengths for the Z-shaped segments and assumed full-active lengths for the lateral devices. The five markers yield a mean K_0 of $45.9\ \text{V}\cdot\text{cm}\cdot\text{aF}/\mu\text{m}$ with $\pm 19\%$ spread. This illustrative coefficient depends on the assumed segment lengths, which were not measured directly.

Under consistent length and bias definitions, $V_\pi L c_j$ is independent of active length, whereas $V_\pi L C_j$ scales with total junction length. The plane compares phase-shifter efficiency at a given

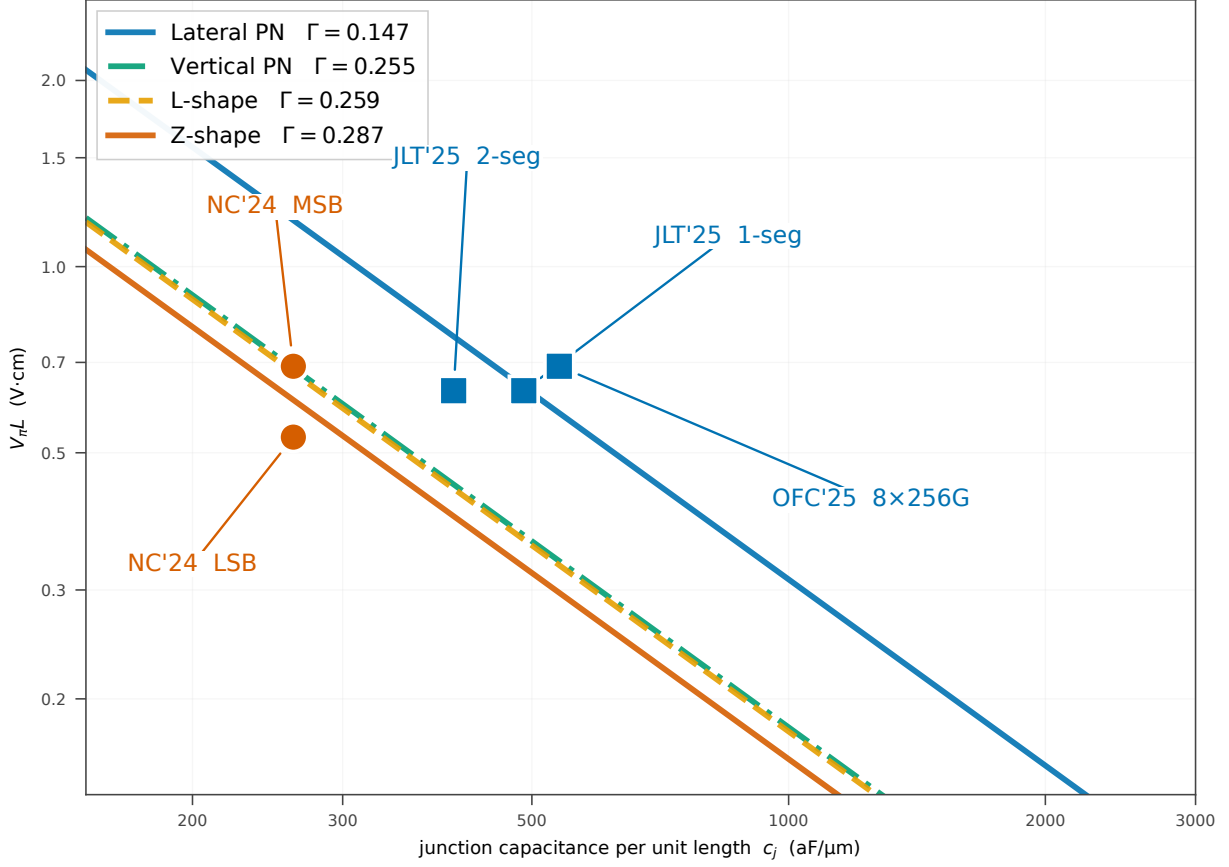


FIG. 5. Efficiency–capacitance comparison under Eq. (18). Lines denote fixed-overlap models; markers denote devices.

capacitance. Loaded electro-optic bandwidth and transmitter quality additionally depend on series resistance, driver and pad loading, and the operating point.

Doping changes capacitance and overlap together, requiring a consistent mode-weighted carrier response in Eq. (18). For the 256G lateral ring in the eight-channel transmitter, an n-type concentration of 10^{19} cm^{-3} and a p-type concentration of $5.3 \times 10^{18} \text{ cm}^{-3}$ give a calculated 38.9 nm depletion width and 20.3 fF junction at -3 V . Its $V_{\pi}L = 0.69 \text{ V}\cdot\text{cm}$ is close to the Z-shaped value of $0.67 \text{ V}\cdot\text{cm}$ with an n-type concentration of $3 \times 10^{18} \text{ cm}^{-3}$ and a p-type concentration of $3 \times 10^{18} \text{ cm}^{-3}$ in the junction region. The Z-shaped junction thus achieves comparable phase efficiency at lower doping through greater overlap.

5. Gain–bandwidth product

At fixed notch depth and normalized detuning, modulation depends on the resonance shift relative to linewidth. For a voltage swing ΔV ,

$$\Delta\lambda_{\text{res}} \simeq \frac{\eta \Delta V Q_L}{\lambda_{\text{res}}} \Delta\lambda_{\text{FWHM}}. \quad (19)$$

The multiplier expresses the shift in linewidths. For the Z-shaped ring, reducing the loaded quality factor Q_L from 3700 to 2500 at 1.6 V per segment requires summed tuning efficiency to rise from 27 to 40 pm/V to preserve that shift. The per-volt factor $\eta Q_L/\lambda_{\text{res}}$ grows with Q_L , while photon-lifetime bandwidth decreases inversely, giving the GBW reference bound within the low-pass approximation of Eq. (17),

$$\text{GBW} = \frac{\eta Q_L}{\lambda_{\text{res}}} f_{\text{EO}} \leq \frac{\eta Q_L}{\lambda_{\text{res}}} f_{\text{ph}} = \frac{\eta c}{\lambda_{\text{res}}^2} \equiv \text{GBW}_{\text{max}}, \quad (20)$$

For the 256G lateral ring in the eight-channel transmitter (Table I), 33 pm/V at 1310 nm gives a GBW reference of 5.8 GHz/V. The measured 67 GHz bandwidth gives 4.2 GHz/V, or 73% of that reference, including electrical loading and detuning-dependent peaking. The modulation-gain GBW requires bandwidth and low-frequency gain at the same detuning (Sec. III D).

Finite junction swing also constrains the photon-lifetime bandwidth when a minimum resonance shift relative to the full linewidth is required. Combining Eq. (19) with $f_{\text{ph}} = c/(\lambda_{\text{res}} Q_L)$ gives

$$f_{\text{ph}} \leq \frac{c\eta\Delta V}{\lambda_{\text{res}}^2 r_{\text{min}}}, \quad (21)$$

where r_{min} is the required resonance shift divided by the full linewidth and ΔV is the delivered junction swing. The bound assumes fixed tuning efficiency and requires r_{min} to come from an independent target such as extinction ratio or OMA, which also depend on coupling and detuning. Taking r_{min} from the same Q_L makes the inequality circular and returns the identity $f_{\text{ph}} = c/(\lambda_{\text{res}} Q_L)$. The bound constrains the photon-lifetime scale alone, whereas the usable electro-optic bandwidth additionally depends on the loaded electrical network and the peaking of the detuned cavity (Sec. III B 6).

6. Loaded electrical network

The lumped estimate $f_{\text{RC,lump}}$ describes an isolated junction charged through R_s . In a driven modulator, the pad, substrate and feed branches also draw current through the source impedance

without charging the junction capacitance C_j , and the feed inductance adds a frequency-dependent series term. An equivalent circuit extracted from measured S_{11} resolves these branches and gives the loaded transfer function H_e of Eq. (22), the fraction of the source swing delivered to the junction at each modulation frequency. That fraction sets the electrical contribution to the electro-optic response and the capacitance optimum of Fig. 8.

A fitted electrical network for the two-segment Z-shaped ring is shown in Fig. 6. It separates the electrical loading of the LSB and MSB segments while both segments modulate the same optical cavity. Fitting the segment S_{11} responses at -3 V gives junction capacitances of 6.6 and 13.2 fF and series resistances of 68.1 and 35.4 Ω , respectively¹². With a 50 Ω source and the fitted pad, feed and substrate parasitics, the networks give junction-voltage bandwidths of 79 and 65 GHz. The S_{11} fit constrains electrical loading alone. The optical decay and the voltage-to-index conversion appear only in the electro-optic S_{21} , measured as detected optical power against electrical drive. At 100 pm laser detuning and -3 V bias, the Z-shaped segments have measured EO bandwidths of 49 and 48 GHz [Fig. 6(c,f)]. Higher detuning increases bandwidth through optical peaking while reducing the low-frequency modulation slope.

Figure 7 applies Eq. (22) to a lateral ring whose published extraction resolves the substrate explicitly³⁸, a different device from the 256G lateral ring of Fig. 8. The junction branch carries $R_s = 246.7$ Ω and $C_j = 10$ fF, the pad $C_{\text{pad}} = 24.6$ fF, and the substrate branch $C_{si} = 107.3$ fF in series with $R_{si} = 833$ Ω . An ideal source gives a junction-voltage bandwidth of 64 GHz, and a 50 Ω source 41 GHz. That 23 GHz difference is the cost of the shunt branches, because current into the pad and substrate flows through the driver without charging C_j . Below its 1.8 GHz crossover the substrate branch is capacitive, and above it the branch approaches the resistive load R_{si} [Fig. 7(b)], so its impedance must be retained when evaluating source loading. Xue *et al.* report 75 GHz for this segment as retained in Table I. The 64 GHz quoted here is the intrinsic pole $1/(2\pi R_s C_j)$ computed from their published element values, so the two numbers rest on different definitions.

Equivalent circuits are not unique. A measured S_{11} constrains the total input impedance, not the internal arrangement of branches, so a network without a substrate branch absorbs that current into C_{pad} and R_s and still reproduces the measurement. The electrical network used by the Z-shaped ring model resolves the feed resistance and inductance, whereas the substrate-resolved lateral-ring network instead resolves the substrate branch. Removing each shunt one at a time and recalculating the junction-voltage bandwidth reveals how sensitive the overall response is to that particular element, while comparing normalized branch-current magnitudes shows how the

current divides among the paths [Fig. 7(d,e)]. With the branches separated, the dominant loading element can be identified before a redesign.

Pad, substrate and junction currents share the feed impedance. Their total admittance reduces the driven-node voltage, followed by the intrinsic junction voltage division,

$$H_e(\omega_m) \equiv \frac{V_j}{V_{\text{src}}} = \underbrace{\frac{1}{1 + Z_{\text{feed}}Y_A}}_{\text{loading}} \cdot \underbrace{\frac{1}{1 + j\omega_m R_s C_j}}_{\text{intrinsic}}, \quad (22)$$

$$Z_{\text{feed}} = R_{\text{src}} + R_p + j\omega_m L_p, \quad Y_A = j\omega_m C_{\text{pad}} + \left(R_{si} + \frac{1}{j\omega_m C_{si}}\right)^{-1} + \left(R_s + \frac{1}{j\omega_m C_j}\right)^{-1},$$

where H_e is the loaded electrical transfer function, ω_m the modulation angular frequency, Z_{feed} the series combination of driver resistance R_{src} , feed resistance R_p and inductance L_p , and Y_A the driven-node admittance. Equation (22) collects the branches resolved by both extractions, taking R_p and L_p from the electrical network of the Z-shaped ring model¹² and the substrate branch C_{si} – R_{si} from the substrate-resolved lateral-ring network³⁸. Each network is the special case of Eq. (22) with its absent branches set to zero. The Z-shaped LSB and MSB segments have extracted feed inductances of 45.2 and 51.1 pH, respectively [Fig. 6(b,e)]. Their feed resistance and inductance remain at zero external source resistance. In the substrate-resolved lateral example, $R_p = L_p = 0$, and an ideal source leaves only the intrinsic junction response [Fig. 7(c)].

7. Design trade-offs above 200 Gb/s

Higher electro-optic bandwidth requires sufficient resonance tuning while limiting the electrical load on the driver. Figure 8 examines this trade-off to identify junction-capacitance targets for higher bandwidth at a given source resistance. These calculated targets guide doping and junction-geometry choices through Eq. (16), with the associated changes in resistance and optical loss evaluated together.

The comparison separates junction capacitance from parasitic loading. The Z-shaped LSB and MSB segments have junction capacitances of 6.6 and 13.2 fF and fitted pad/via capacitances of 31.6 and 33.1 fF, respectively [Fig. 6(b,e)]¹². For the 256G lateral ring at -3 V, the S_{11} fit gives a node capacitance near 60 fF³⁶. Below the reported 167 GHz electrical pole, the measured reflection remains predominantly capacitive, constraining node capacitance more tightly than series resistance. The calculated junction capacitance is 20.3 fF, leaving approximately 40 fF attributed to pad and routing.

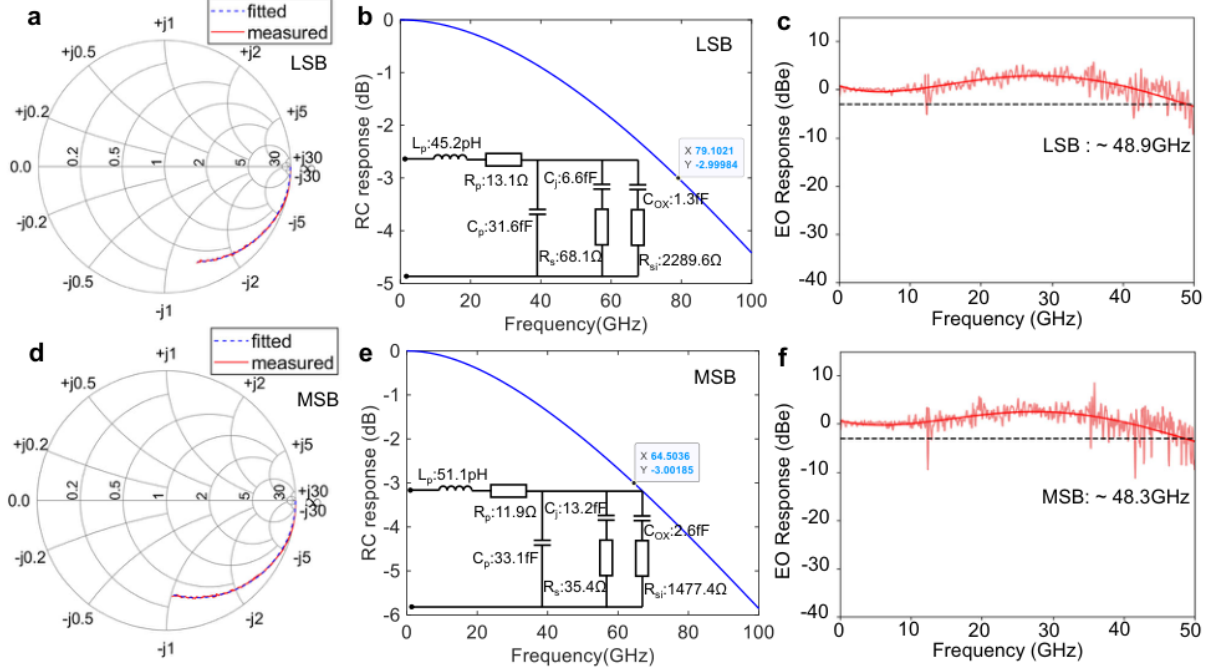


FIG. 6. Z-shaped ring characterization at -3 V. **(a,d)** Measured/fitted S_{11} . **(b,e)** Extracted circuits and calculated responses. **(c,f)** Measured EO responses. Cropped/rescaled from Yuan *et al.*, Fig. 3¹², CC BY 4.0.

The solid curves in Fig. 8 combine the loaded electrical response $H_e(\omega_m)$ with the cavity response $H_o(\omega_m)$. At each operating point, H_o describes small-signal resonance modulation with intrinsic loss and coupling held fixed and is normalized to unity at zero frequency. The combined response is

$$H_{EO}(\omega_m) = H_e(\omega_m) H_o(\omega_m), \quad (23)$$

and f_{EO} is the first frequency at which $|H_{EO}|^2$ falls 3 dB below its DC value. An effective de-tuning is fitted to each reference electro-optic bandwidth at a $50\ \Omega$ source resistance, with the electrical parameters held fixed. This calibration retains cavity peaking, which is omitted from the approximate low-pass budget of Eq. (17).

Figure 8 uses total junction capacitance C_j because source loading depends on the individual junction and parasitic branches. The Z-shaped LSB and MSB segments share a cross-section and capacitance per unit length, but their total junction capacitances differ by a factor of two. Their separate electrical networks must therefore be retained when comparing capacitance optima. Under inverse-length resistance scaling, the intrinsic product $R_s C_j = \rho_s c_j$ is length-independent, where $\rho_s = R_s L_j$.

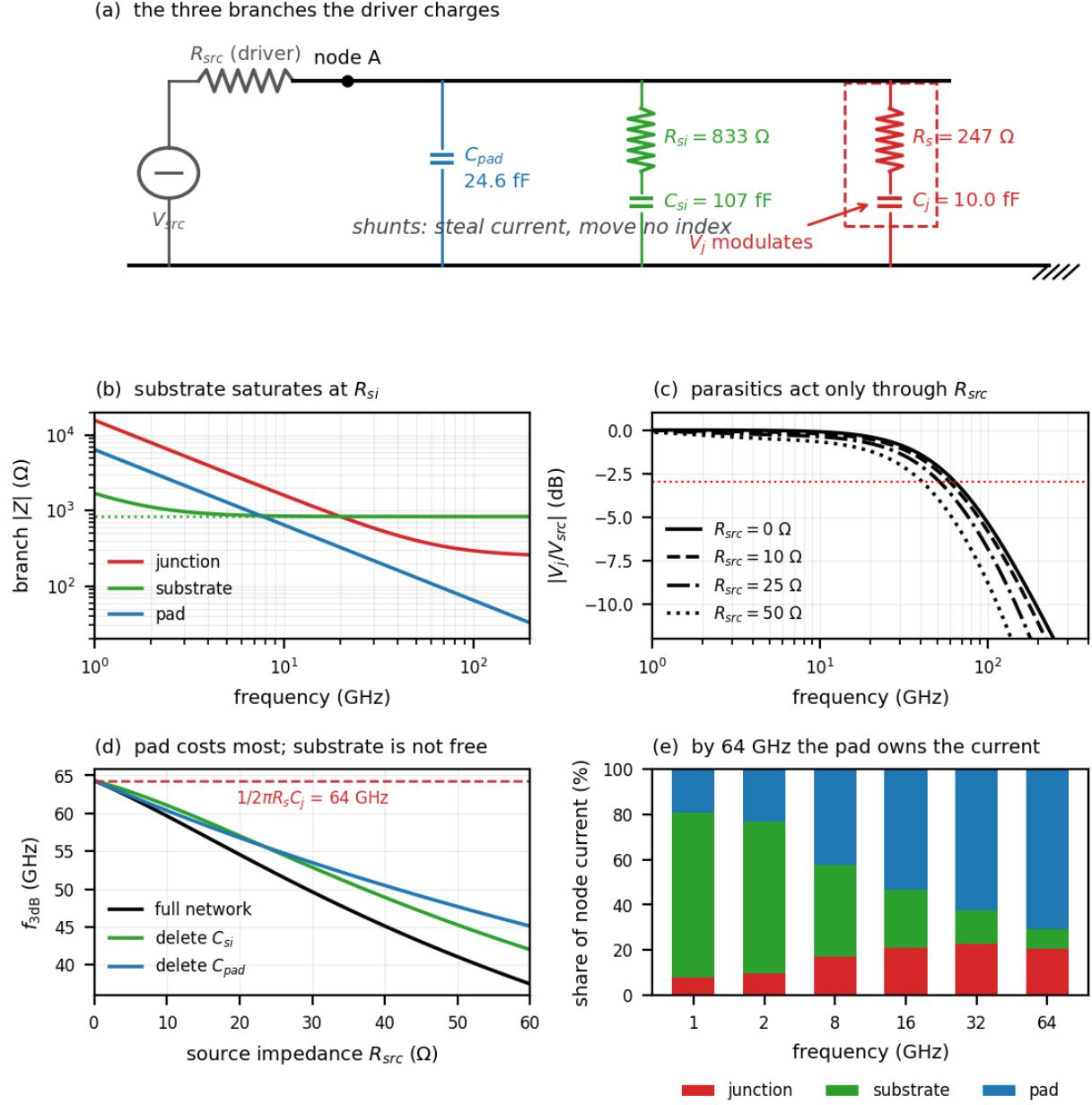


FIG. 7. Substrate-resolved lateral loading model³⁸. (a) Circuit. (b) Branch impedances. (c) Source-dependent junction response. (d) Bandwidth with individual shunts removed. (e) Branch-current magnitudes normalized by their sum.

The capacitance sweep assumes fixed overlap and active length. Increasing C_j then increases the charge displaced per volt and improves tuning efficiency. To preserve the resonance shift relative to linewidth at the same junction swing, the cavity is redesigned with a lower loaded quality factor. This broadens the photon linewidth while the increased capacitance slows junction

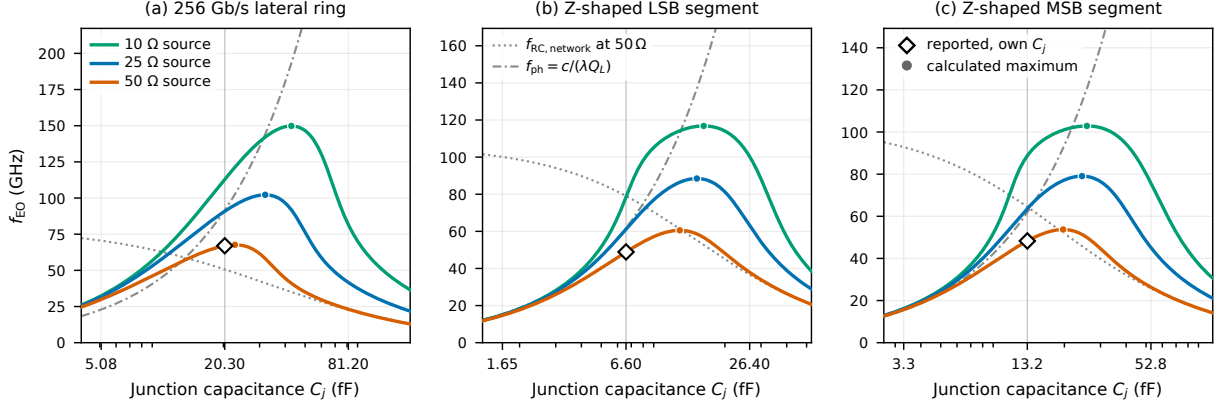


FIG. 8. Electro-optic bandwidth against junction capacitance for the **(a)** 256G lateral ring, **(b)** Z-shaped LSB and **(c)** MSB segments. Solid curves are the cascade at 10, 25 and 50 Ω source resistance. Grey curves show $f_{RC, network}$ at 50 Ω (dotted) and f_{ph} (dash-dotted). Circles mark calculated maxima and diamonds measured bandwidths. Optima are conditional on the stated scaling assumptions and fitted detunings.

charging. Series resistance, pad, feed and substrate parameters remain fixed during the sweep.

At fixed wavelength, geometry, overlap Γ and carrier-response factor K_0 , the product $(V_\pi L)c_j \simeq K_0/\Gamma$ is constant, giving $\eta = FSR L_j/[2(V_\pi L)] \propto c_j$. Maintaining the same relative resonance shift and junction swing ΔV gives the following scalings, with junction-dominated electrical loading and fixed ρ_s ,

$$Q_L = \frac{m\lambda_{res}}{\eta\Delta V} \propto c_j^{-1}, \quad f_{ph} = \frac{1}{2\pi\tau_p} = \frac{c}{\lambda_{res}Q_L} \propto c_j, \quad f_{RC, lump} \propto c_j^{-1}. \quad (24)$$

where m is the resonance shift divided by the full linewidth. These assumptions make the calculated optima conditional design targets. A doping change must also be evaluated for its effects on resistance, optical overlap and absorption.

For the assumed active lengths of 39.5 μm for the Z-shaped ring and 37.7 μm for the 256G lateral ring, total capacitance follows $C_j = c_j L_j$. Increasing source resistance from 10 to 50 Ω shifts the optimum capacitance from 25.7 to 19.7 fF for the Z-shaped ring and from 42.9 to 22.8 fF for the 256G lateral ring. The corresponding peak bandwidths fall from 103 to 54 GHz and from 150 to 68 GHz. The estimated 20.3 fF junction of the 256G lateral ring is already near its 50 Ω optimum. Within this model, reducing source resistance offers more bandwidth improvement than further reducing junction capacitance.

Figure 9 separates the electrical and optical contributions in the frequency domain. For the

256G lateral ring [Fig. 9(a)], the loaded electrical cutoff is 51 GHz (dotted curve), yet the composite electro-optic response reaches 67 GHz (solid curve). This bandwidth extension occurs because optical cavity peaking compensates for the high-frequency electrical roll-off. Modeled cavity peaking increases to approximately 3.5 dB at $|x| = 1.13$ and 9.6 dB at $|x| = 2$ (dashed curves). At the fitted operating detuning ($|x| = 0.90$), approximately 2 dB of optical peaking near 48 GHz acts as an analog pre-emphasis filter that lifts the attenuated electrical signal, flattening the cascaded response $H_{EO} = H_e H_o$ and extending the net 3 dB bandwidth to 67 GHz with less than 0.2 dB of passband ripple.

This detuning of $|x| = 0.90$ is a calibrated fit selected to match the measured 67 GHz bandwidth under the published zero-inductance lumped network ($L_p = 0$), rather than a directly measured operating point. Xue *et al.* report operating at the 3 dB through-port insertion loss point, corresponding to $|x| \approx 0.98$, which yields a calculated bandwidth of 72.5 GHz under this idealized lumped model. If on-chip routing instead introduces a modest parasitic inductance of $L_p = 20$ pH, the loaded electrical cutoff rises from 51 to 58 GHz, so reaching 67 GHz requires less cavity peaking and shifts the matching detuning inward to $|x| \approx 0.80$, near the maximum dynamic OMA point. These variations illustrate that extracting junction dynamics and predicting composite bandwidth require precise knowledge of the laser parking offset relative to the hot resonance.

For the Z-shaped ring [Fig. 9(b,c)], the loaded electrical cutoffs are 79 GHz for the LSB segment and 65 GHz for the MSB segment. Because the Z-shaped device operates closer to the notch ($|x| \approx 0.58\text{--}0.62$) to maximize optical modulation amplitude, cavity peaking remains below 1 dB, leaving the electro-optic bandwidths (49 and 48 GHz) below the electrical cutoffs. The solid electro-optic responses use the same networks and fitted detunings as Fig. 8, including the parasitics extracted for the Z-shaped ring and the assumed junction/pad partition of the 256G lateral ring.

C. Optical nonlinearities in the cavity

Absorption of the circulating optical field heats the waveguide and increases its effective index. The resulting resonance redshift changes the laser detuning, modulation slope and OMA⁵². Thermal control must therefore maintain the operating point as optical power and temperature vary. At

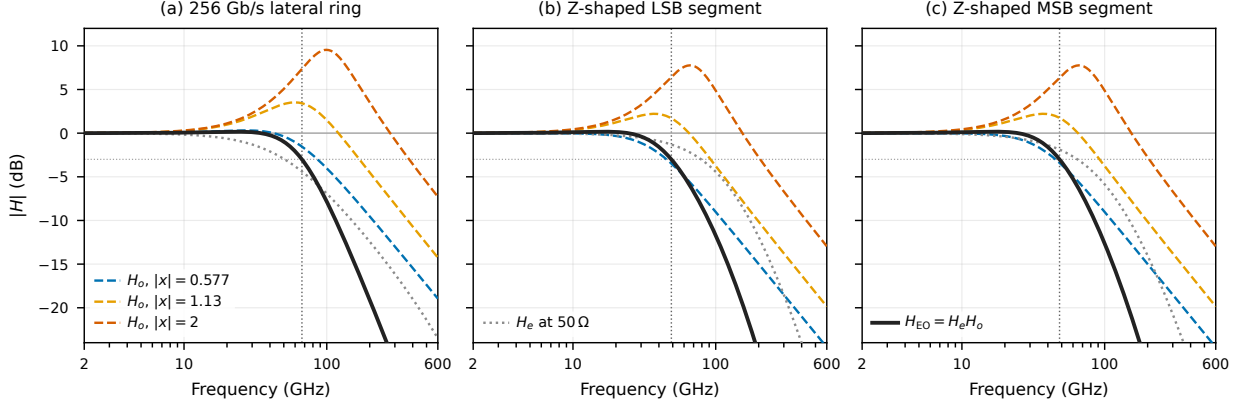


FIG. 9. Frequency responses for the **(a)** 256G lateral ring, **(b)** Z-shaped LSB and **(c)** MSB segments. Dashed curves show H_o at three detunings, dotted curves show H_e at 50Ω , and solid curves show H_{EO} at the fitted detuning and the same source resistance. Vertical lines mark measured electro-optic bandwidths.

fixed bias, a small temperature change gives

$$\Delta\lambda_{\text{res}} = \frac{\lambda_{\text{res}}}{n_g} \frac{\partial n_{\text{eff}}}{\partial T} \Delta T. \quad (25)$$

where ΔT is the effective temperature change sampled by the optical mode, with thermal expansion neglected.

At thermal equilibrium, the optical-heating contribution follows the average absorbed power⁴⁴,

$$\bar{P}_{\text{abs}} = \langle P_{\text{abs}}(t) \rangle, \quad \Delta\lambda_{\text{res}} = \left. \frac{\Delta\lambda_{\text{res}}}{\bar{P}_{\text{abs}}} \right|_{\text{ref}} \bar{P}_{\text{abs}}, \quad (26)$$

where P_{abs} is the thermalized optical power. The reference ratio is evaluated at nonzero absorbed power, with the resonance shift measured from zero optical heating at fixed bias, heater power and package temperature.

For continuous-wave (CW) input at fixed bus power, intrinsic loss and external coupling, constant thermal sensitivity gives

$$\Delta\lambda_{\text{res}} = \frac{\Delta\lambda_{\text{res}}|_{x=0}}{1+x^2}. \quad (27)$$

Here $\Delta\lambda_{\text{res}}|_{x=0}$ is the on-resonance thermal shift at the same bus power. The detuning x is measured from the hot resonance, whose position depends on absorption. Absorption and resonance shift must therefore be solved self-consistently. Through-port transmission follows Eq. (12), including off-resonance loss. An assumed peak absorbed fraction of approximately 84% and thermal sensitivity of 160 pm per milliwatt absorbed give an on-resonance shift of 134 pm per milliwatt of bus power. Heater tuning³⁶ enters the combined wavelength balance in Sec. IV D.

For a blue-detuned laser ($\lambda < \lambda_{\text{res}}$), heating moves the resonance farther from the laser, reducing absorption and opposing further heating. With red detuning, the resonance initially moves toward the laser, increasing absorption and reinforcing heating. This feedback distorts the transmission recorded during a slow laser-wavelength scan and can produce multiple stable thermal states. In the model of Sec. V, absorbed power drives a thermal wavelength-shift state that updates the cavity detuning. The spectra and eyes in Fig. 11 describe modulation about the settled hot resonance.

Two-photon absorption also generates electron–hole pairs at a rate proportional to the squared local optical intensity. The accumulated carriers lower the effective index through free-carrier dispersion (FCD), shifting the resonance blue through Eq. (13). Free-carrier absorption shortens the intrinsic lifetime and changes the notch depth and linewidth. Reverse bias sweeps carriers out, reducing their residence time and both effects⁵². The Z-shaped ring combines this carrier extraction with two-segment modulation.

Photocurrent can additionally change the operating bias. Current through the DC feed resistance produces a voltage drop that reduces the junction’s reverse bias. The depletion region contracts, increasing the carrier population sampled by the optical mode and producing a further blue shift. With 33 pm/V tuning, 100 μA through 1 k Ω gives approximately 3 pm of shift. Low-resistance feeds help preserve junction bias and the relative modulation weights of the Z-shaped segments.

Figure 10(a) compares separate thermal and carrier contributions using a 1310 nm reference and $Q_L = 3700$, corresponding to the Z-shaped ring’s 354 pm linewidth. The assumed thermal shift reaches one linewidth at 2.7 mW bus input. The carrier curves use reference carrier-density and mode-volume parameters from the 256G lateral-ring model, with residence times of 10 ps with sweep-out and 1 ns without it. These curves illustrate the individual mechanisms rather than a coupled thermal–carrier solution for the Z-shaped ring.

At 5 mW bus input, the assumed thermal redshift is 668 pm, compared with reference FCD blue shifts of 5.7 pm at 1 ns residence time and 0.1 pm at 10 ps. Faster carrier extraction reduces the blue shift, while heating dominates the resonance displacement under these assumptions. The thermal shift therefore provides the main contribution to the required wavelength-compensation range in this comparison.

Panel (b) shows calculated CW through-port transmission versus laser wavelength at fixed bus powers of 0, 1 and 5 mW. The model retains $Q_L = 3700$ and the same thermal sensitivity, but sets both peak absorption and notch depth to 97%, with an unheated resonance at 1310 nm. The 0 mW

curve denotes the weak-probe limit. The horizontal axis is the scanned laser wavelength, while the ring resonance changes with the self-consistent thermal state.

Above the calculated 1.8 mW bistability threshold, the equilibrium transmission curve folds back over a range of laser wavelengths. Within that range, one laser wavelength can support two stable hot-resonance positions separated by an unstable state. During a slow scan, transmission follows a stable branch until that branch ends, then jumps to another branch. The jump occurs at different wavelengths for opposite scan directions, producing hysteresis. Laser and heater acquisition must therefore reach the intended operating branch.

The threshold scales with linewidth and inversely with peak absorbed fraction and thermal sensitivity. Reducing Q_L from 3700 to 2500 raises the calculated threshold to 2.6 mW when the other assumptions are unchanged. This threshold marks the onset of possible hysteresis during acquisition. Blue-detuned equilibria remain locally stable above it in the thermal-only model because heating reduces absorption.

The operating-power example in Sec. IV A uses 11.3 mW bus input for the 256G lateral ring. At the selected hot detuning, approximately 5.8 mW is absorbed, giving a 920 pm redshift spanning 1.8 linewidths. Cold resonance placement and heater reserve are chosen together to maintain that operating point. The 15 mW bus-power cap in Sec. III D 2 is an analysis assumption. Establishing a device power limit additionally requires checking tuning range, photocurrent-induced bias changes, nonlinear absorption, dynamic stability and eye quality at the intended operating conditions.

D. Laser detuning and the operating point

1. Laser detuning

Laser detuning determines how efficiently a resonance shift produces optical modulation and how the cavity responds to rapid voltage changes. The operating point must therefore balance modulation amplitude, bandwidth and waveform distortion.

In the depletion-width model of Fig. 11, increasing reverse bias expands the depletion region, raising the effective index and reducing absorption. At fixed laser wavelength, the voltage-dependent resonance position and linewidth determine the detuning through Eq. (12). The detuning and notch depth at each voltage level then determine the settled optical power. Integrating

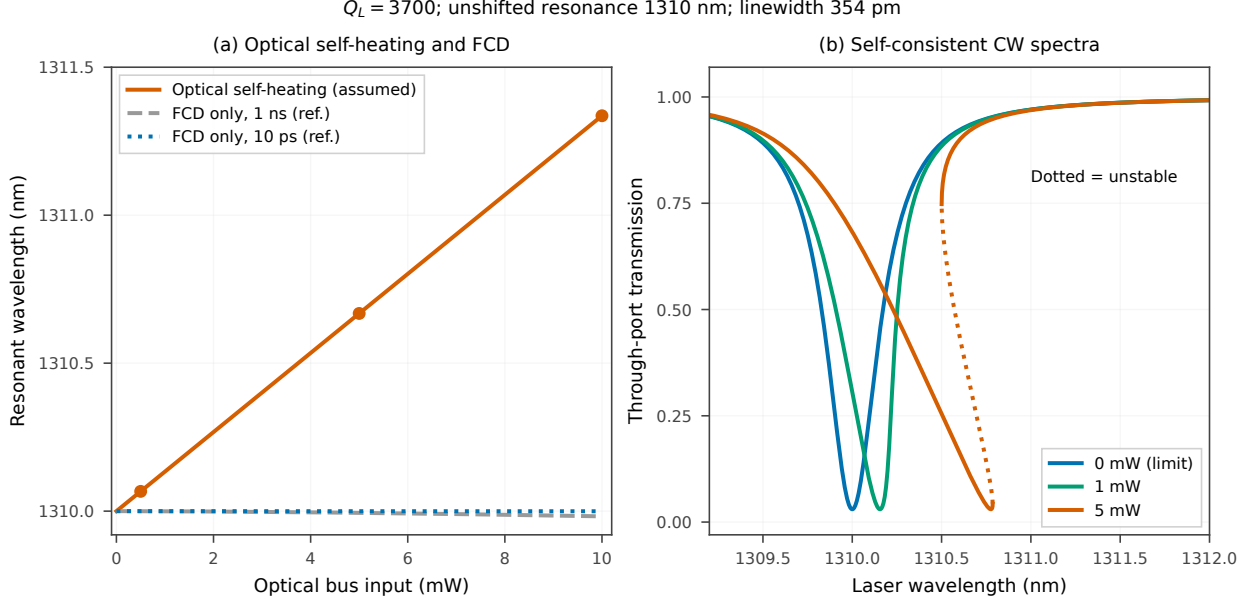


FIG. 10. Calculated power-dependent response at $Q_L = 3700$. **(a)** Separate thermal-only and FCD-only resonance wavelengths versus bus power. **(b)** CW transmission equilibria versus laser wavelength, stable (solid) and unstable (dotted). The 0 mW curve is the weak-probe limit.

Eq. (1) with the filtered junction voltage includes cavity memory and interference with the directly transmitted field (Sec. V). Beyond the maximum-slope operating point, increasing detuning can broaden the optical response through peaking while reducing low-frequency modulation gain.

At 1310 nm, unit normalized detuning corresponds to half a linewidth, or 177 pm for the Z-shaped ring at $Q_L = 3700$ and 262 pm for the 256G lateral ring at $Q_L = 2500$. Their respective extreme-state resonance shifts of 44 and 59 pm span approximately 0.25 and 0.23 in normalized detuning [Fig. 12].

2. Maximum slope, optical modulation amplitude and gain-bandwidth

Maximum slope, maximum optical modulation amplitude (OMA) and maximum gain-bandwidth product (GBW) describe different operating objectives. For a fixed-depth Lorentzian, maximum slope occurs at $|x| \simeq 0.58$, where zero local curvature suppresses second-order static distortion⁵³. The static finite-swing OMA, defined by the outer settled powers, peaks nearby for the fixed-loss 256G lateral-ring model at $Q_L = 2500$ and 1.8 V_{pp}. At 128 GBaud, cavity dynamics move the waveform-extrema OMA maximum to $|x| \simeq 0.80$ [Table II]. Waveform extrema include transition

TABLE II. Fixed-loss bias comparison at $Q_L = 2500$, 128 GBaud and $1.8 V_{pp}$.

Objective	$ x $	Offset (pm)	Mean IL (dB)	k_{OMA} (%)	$P_{bus,req}$ (mW)
Local slope	0.58	151	5.9	16	11.9
Static finite-swing OMA	0.58	153	5.8	16	11.9
Dynamic waveform OMA	~ 0.80	210	4.0	17	11.3
GBW with 167 GHz RC	~ 0.93	244	3.3	17	11.5
Bare-cavity GBW	~ 1.14	299	2.5	16	12.1

overshoot, whereas the sampled eye opening depends on adjacent-level separation at the decision instant.

The detuning-dependent GBW combines low-frequency modulation gain with the 3 dB bandwidth^{53,54}. This definition differs from the resonance-shift metric in Eq. (20). Electrical filtering reduces the high-frequency enhancement, while further detuning lowers the low-frequency gain. With the assumed 167 GHz electrical pole, the grid-sampled GBW optimum shifts from the bare-cavity value $|x| = 1.14$ to $|x| = 0.93$ [Table II].

Voltage-dependent absorption changes these optima because the linewidth and notch depth vary during modulation. With fixed external coupling, the loss-aware 256G lateral-ring model gives a notch extinction ratio (ER) of 19.6–20.5 dB across the drive swing [Fig. 11]. The maximum-slope, waveform-OMA and GBW operating points occur at $x = -0.59$, -0.80 and -0.98 , respectively. The reference model uses a 524 pm linewidth, coupling ratio $\rho = 1.2$ at -3 V and tuning efficiency of 33 pm/V. An n-type concentration of 10^{19} cm^{-3} , a p-type concentration of $5.3 \times 10^{18} \text{ cm}^{-3}$ and an assumed 350 nm absorbing width determine the undepleted-region loss. Of the reference 151 dB/cm loss, 31 dB/cm remains fixed.

The conversion factor k_{OMA} is the waveform-extrema OMA divided by ring-bus input power. Mean insertion loss (IL) is calculated from the mean linear power of the same record, excluding coupling and routing losses. The powers in Table II assume 0.6 mW receiver OMA and 5 dB downstream loss (Sec. IV A). Compared with maximum waveform OMA, the RC-inclusive GBW operating point reduces mean insertion loss by 0.7 dB but also reduces modulation contrast, leaving a 0.063 dB increase in required power. Under the continuous-wave (CW) assumptions of Sec. III C, absorbed power falls from 5.8 to 5.1 mW, reducing thermal redshift.

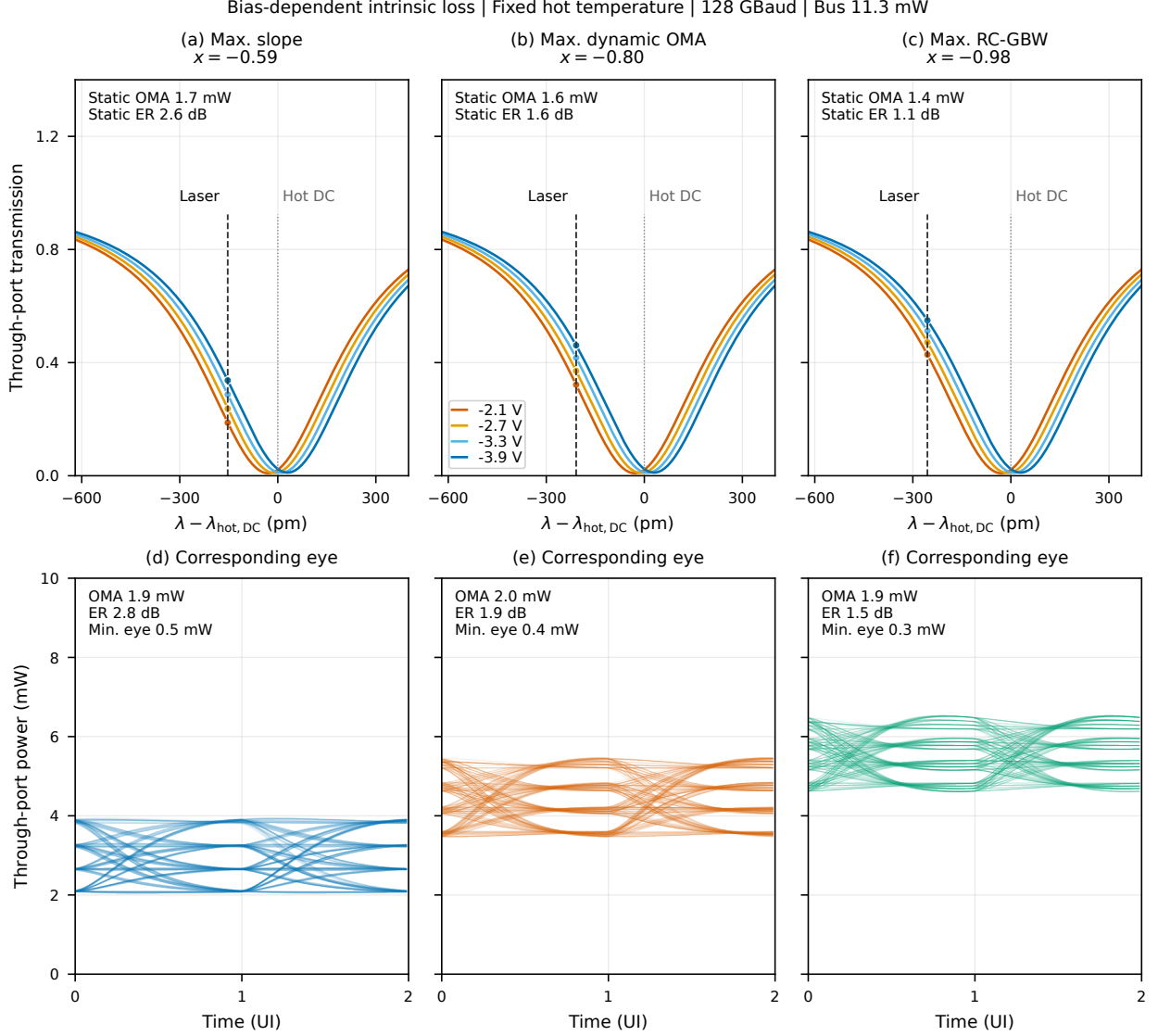


FIG. 11. Calculated spectra (a–c) and noise-free PAM4 eyes (d–f) at maximum slope, waveform OMA and RC-inclusive GBW, respectively. Bias-dependent loss, 128 GBaud, 11.3 mW bus input, 1.8 V_{pp} and a 167 GHz drive pole are used. Dashed lines mark the laser; markers denote settled levels.

For our Z-shaped ring, the overcoupled approximation in Fig. 12(a,c) uses the Z-shaped LSB segment's 79 GHz electrical pole and a combined tuning efficiency of 27 pm/V, with both segments perturbed together. The calculated GBW maximum is near $|x| = 0.92$. At 100 GBaud and 1.6 V_{pp} per non-return-to-zero (NRZ) segment, waveform OMA peaks near $|x| = 0.62$ and reaches 16% of bus input power, using an assumed 9/18 pm/V tuning split [Fig. 12(b)]. The undercoupled model used for the noise comparison retains both fitted segment networks and gives a GBW optimum near $|x| = 0.97$ (Sec. V).

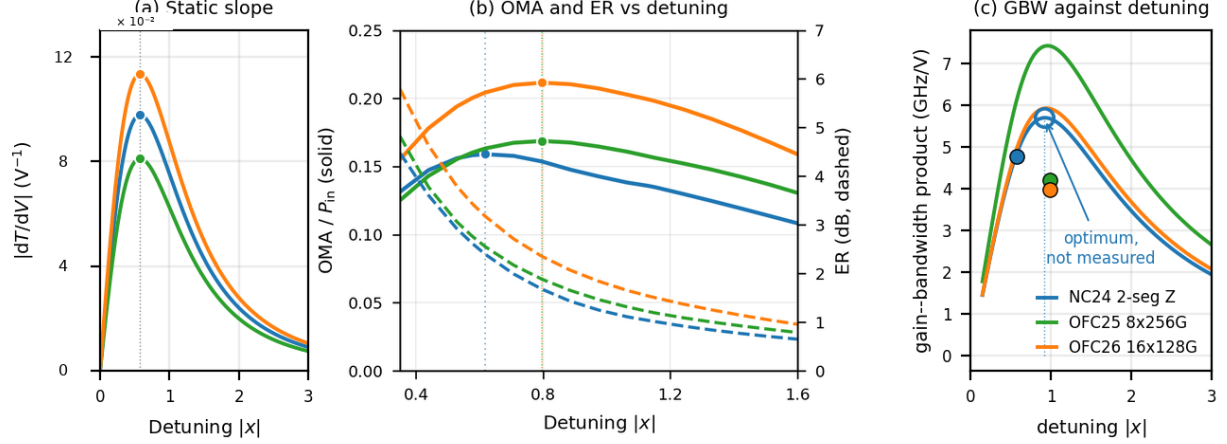


FIG. 12. Detuning-dependent response. **(a)** Static modulation slope. **(b)** Simulated waveform OMA (solid) and ER (dashed). **(c)** GBW from measured electro-optic bandwidths (filled markers) and the predicted Z-shaped ring optimum (open). Device-specific rates and drives are specified in the text.

For the 128 Gb/s lateral ring³⁷, the model gives a normalized waveform OMA of 21% near $|x| = 0.80$ at 64 GBaud. The calculation uses $Q_L = 4300$, 1.6 V_{pp} , 27 pm/V tuning, an 18 dB overcoupled notch and an estimated 84 GHz electrical pole. This larger OMA is obtained at a lower baud rate and with a narrower resonance than in the 256G lateral-ring model. All waveform OMA and ER values in Fig. 12(b) use noise-free extrema before receiver processing. Power normalization in the plots and Table IV refers to the ring bus after input-coupling loss.

Receiver filtering and equalization determine whether these operating-point advantages translate into lower required power or a higher supported baud rate. Figure 13(a,b) compares the operating points of the fixed-loss 256G lateral-ring model, retaining $Q_L = 2500$, coupling ratio $\rho = 1.22$, 33 pm/V tuning and 1.8 V_{pp} . This comparison uses the refined GBW bias $x = -0.96$. The electrical pole and Bessel receiver bandwidth remain at 167 and 64 GHz as baud rate increases, and transmitter feed-forward equalization (FFE) is bypassed.

The sampling phase, decision thresholds and optional receiver equalizer are determined from training data. Gaussian noise is then applied statistically to held-out decision samples to calculate the symbol error rate (SER),

$$p_{e,i}(\sigma) = \mathcal{Q}\left(\frac{y_i - b_{\ell_i}}{\sigma}\right) + \mathcal{Q}\left(\frac{b_{\ell_i+1} - y_i}{\sigma}\right), \quad \text{SER}(\sigma) = \frac{1}{4} \sum_{\ell=0}^3 \frac{1}{N_{\ell}} \sum_{i: \ell_i=\ell} p_{e,i}(\sigma), \quad (28)$$

where y_i is the noise-free decision sample, ℓ_i its transmitted level, b_{ℓ} the decision boundary, σ the root-mean-square (RMS) noise amplitude and N_{ℓ} the number of samples at level ℓ . The function

$\mathcal{Q}$ is the standard-normal upper tail, with $b_0 = -\infty$ and $b_4 = +\infty$. Errors are averaged with equal weight for the four transmitted levels. The target is $1.5\mathcal{Q}(3.41) \simeq 4.8 \times 10^{-4}$.

The propagated relative intensity noise (RIN) variance is subtracted from the total variance allowed at the target SER. The positive remainder determines the additive RMS noise margin $M(P_0)$, referred to the input of the receiver equalizer with downstream attenuation factored out. At fixed detuning, linear optical-power scaling gives the required bus and laser powers,

$$P_{\text{bus,req}} = P_0 \frac{\sigma_{\text{add}}}{10^{-L_{\text{down}}/10} M(P_0)}, \quad P_{\text{laser,req}} = P_{\text{bus,req}} 10^{L_{\text{in}}/10}, \quad (29)$$

where P_0 is the reference bus power, σ_{add} the fixed optical-power-equivalent receiver noise before equalization, and L_{down} and L_{in} the downstream and input losses in dB.

The additive noise of 21.4 μW RMS is calibrated at the maximum-OMA bias with receiver FFE, 11.3 mW bus power and 128 GBaud. The calculation assumes 5 dB downstream loss, -155 dB/Hz input RIN over 0–64 GHz and 2.5 dB input loss, with no additional margin. Shot noise, jitter, dispersion and forward-error correction are excluded. Residual intersymbol interference (ISI) is measured as the RMS spread about each level mean, normalized by the outer-level centroid separation. Power and ISI comparisons use 128 GBaud. The highest supported rates satisfy the SER target on both tested symbol records within an assumed 15 mW bus-power cap.

For the reference record at 128 GBaud, the maximum-slope, OMA and GBW operating points require 17.5, 12.2 and 13.0 mW without receiver equalization, and 11.7, 11.3 and 12.1 mW with five-tap receiver FFE, respectively. Maximum OMA requires the least power in both cases. The highest supported rate has a different ranking. Without equalization, the lower ISI at the GBW operating point supports the highest rate within the 15 mW cap. Receiver FFE reverses this ranking, allowing the maximum-OMA operating point to support the highest rate [Fig. 13(a,b)].

Maximum slope produces the least overshoot [Fig. 13(c,d)] but the most ISI before equalization. After equalization, its smaller RMS level spread still gives less noise margin than the maximum-OMA operating point because the absolute separation between levels is also smaller. Operating-point selection therefore requires both residual distortion and optical level separation to be evaluated with the intended receiver.

3. Chirp

Voltage-induced resonance shifts change both the through-port optical power and phase in Eq. (1). The time derivative of phase relative to the laser gives the instantaneous frequency excu-

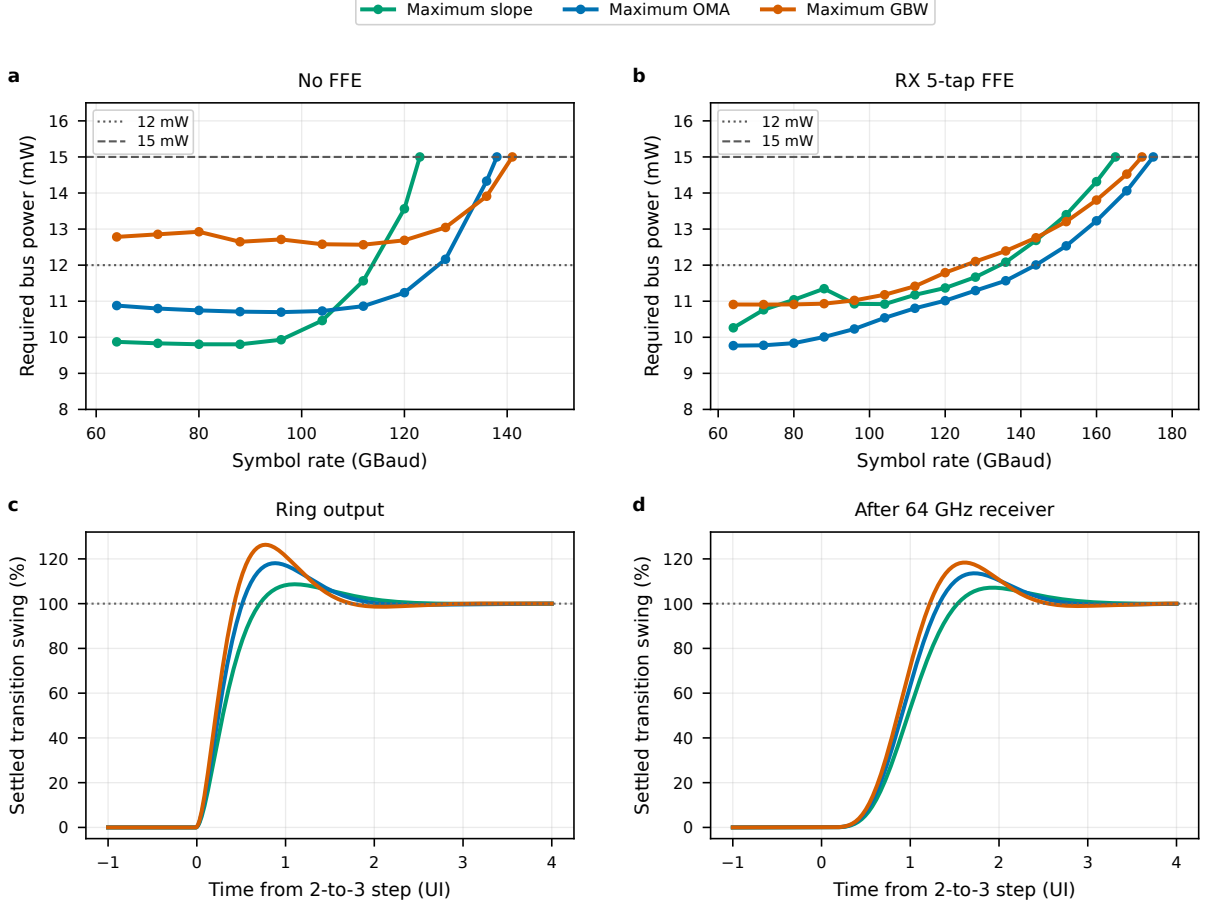


FIG. 13. Fixed-loss operating-point comparison. **(a,b)** Required bus power versus symbol rate without and with a five-tap receiver feed-forward equalizer; each trace stops at the highest rate meeting the 4.8×10^{-4} symbol-error target within 15 mW. Dotted and dashed lines mark 12 and 15 mW. **(c,d)** Ring and receiver 2-to-3 steps at 128 GBaud. Transmitter equalization is bypassed and the receiver is held at 64 GHz.

sion, or chirp. For the fixed-loss 256G lateral-ring model at $Q_L = 2500$, 1310 nm, 128 GBaud, 1.8 V_{pp} and hot detuning $x = -0.80$, the calculated RMS chirp is 1500 MHz.

Chirp affects the detected intensity when the channel introduces frequency-dependent delay or unequal sideband transmission. A chirped waveform and a constant-phase control with identical transmitted intensity produce the same detected waveform through a flat channel. Their RMS difference increases to 1% after an assumed integrated dispersion of 2.5 ps/nm and to 3% after an assumed single-pole optical bandpass filter with a 120 GHz power full width at half maximum, centered 30 GHz above the laser. Both comparisons use a periodic 256-symbol PAM4 record, identical timing and a fourth-order 64 GHz Bessel receiver without equalization or added noise.

The differences are normalized to the flat-channel detected peak-to-peak excursion.

IV. THERMAL TUNING AND RESONANCE PLACEMENT

At 128 GBaud PAM4, the selected $x = -0.80$ bias minimizes modeled bus power among the fixed-loss candidates at equal receiver OMA (Sec. III D), placing the laser 210 pm blue of the 256G lateral ring’s hot resonance. Self-heating changes the alignment after turn-on, coupling optical power to cold resonance placement and heater control.

A. Operating optical power

Using the waveform-extrema conversion k_{OMA} defined in Sec. III D, the required ring-input bus power is

$$P_{\text{bus,dBm}} = S_{\text{RX}} + L_{\text{post}} + M - 10\log_{10} k_{\text{OMA}}, \quad (30)$$

where S_{RX} is the required receiver outer OMA in dBm, L_{post} the downstream loss and M the engineering margin in dB. The factor $-10\log_{10} k_{\text{OMA}}$ represents the electro-optic modulation conversion penalty of the microring. Because reverse bias shifts the resonance over only a fraction of its linewidth ($\Delta x \approx 0.23$), the optical transmission swings between $T_0 \approx 0.38$ and $T_3 \approx 0.55$, converting approximately 16.8% of the input continuous-wave power into peak-to-peak modulated outer OMA. This 7.7–8.0 dB conversion penalty is distinct from the receiver’s $10\log_{10} 3 \simeq 4.8$ dB PAM4 inner-eye slicing penalty, which arises from subdividing the outer optical swing into three equal decision eyes at the slicer.

The IEEE 200G optical receiver baseline at 106.25 GBaud PAM4 (212.5 Gb/s gross) specifies $S_{\text{RX}} = -3.2$ dBm for low TECQ²¹. Table III summarizes the optical link budget under co-packaged edge coupling, with 1.0 dB input and output coupler losses. Downstream passive loss totals $L_{\text{post}} = 3.5$ dB, comprising 1.5 dB on-chip routing, 1.0 dB output edge coupling and 1.0 dB external fiber and connector loss. At the nominal 11.3 mW bus power baseline (+10.5 dBm), the link delivers -0.7 dBm outer OMA to the receiver for the 256G lateral ring and -1.0 dBm for the Z-shaped ring, providing healthy unallocated implementation margins of 2.5 and 2.2 dB, respectively. With 1.0 dB input edge coupling, the continuous-wave laser output requirement is 14.2 mW (+11.5 dBm) per lane, comfortably within commercial DFB laser ratings.

TABLE III. Per-lane optical link budget with 1.0 dB edge coupling at 11.3 mW bus power.

Link stage (laser $\rightarrow$ receiver)	Loss (dB)	256G Lateral (dBm)	Z-shaped (dBm)
Laser source output	—	+11.5	+11.5
Input edge coupler	1.0	+10.5	+10.5
CW bus power ($P_{\text{bus}} = 11.3$ mW)	—	+10.5	+10.5
Modulator OMA conversion (k_{OMA})	7.7 / 8.0	+2.8	+2.5
On-chip routing	1.5	+1.3	+1.0
Output edge coupler	1.0	+0.3	0.0
External fiber channel	1.0	−0.7	−1.0
Received outer OMA at TP3	—	−0.7	−1.0
Receiver sensitivity (S_{RX})	—	−3.2	−3.2
Implementation margin (M)	—	+2.5 dB	+2.2 dB

B. Heating at the selected operating point

At 11.3 mW bus input and held $x = -0.80$, the assumed 51% CW absorbed fraction gives 5.8 mW absorption and a 920 pm redshift. The shift spans 1.8 linewidths, far exceeding the 59 pm drive excursion. Under modulation, average absorption determines the thermal shift through Eq. (26), while the shifted resonance changes absorption through Eq. (27). The hot state must therefore be determined self-consistently.

The comparison of our Z-shaped ring and the 256G lateral ring in Fig. 14 uses the same 11.3 mW nominal bus power for both rings. Both models assume 160 pm per milliwatt absorbed and 54.6 pm per kelvin, with $\bar{P}_{\text{abs}} \simeq 0.84P_{\text{bus}}/(1+x^2)$. Cold placement and heater power are jointly optimized for hot $x = -0.80$ over 10.2–12.4 mW bus power, 40–50 °C package temperature and ± 100 pm fabrication error, with 0.5 mW reserve at each heater limit. The assumed heater-power range for the 256G lateral ring is 0–8 mW. The Z-shaped ring’s heater-power range is 0–32 mW, corresponding to the demonstrated 0–4 V sweep at 500 Ω ¹².

At 1310 nm, the models of the Z-shaped ring and the 256G lateral ring use $Q_L = 3700$ and 2500, with 16 dB notch ER assigned to the Z-shaped ring’s -3 V reference. Their target resonances lie 142 and 210 pm red of the laser. The same 920 pm self-shift spans 2.6 linewidths of the Z-shaped ring but 1.8 of the 256G lateral ring, giving different feedback branches despite equal

absorbed power. Optimized cold resonances about 2.5 nm blue of the laser at 0 V and 25 °C require minimum nominal heater powers of 6.9 and 3.4 mW, respectively, with corner ranges of 0.5–13.2 and 0.5–6.3 mW.

For the 256G lateral ring, the hottest, highest-power and redmost fabrication corner sets the lower heater reserve, limiting how far the cold resonance can move red to reduce nominal heating. Laser turn-on with the heater off gives only 70 pm self-shift. Heater tuning moves the resonance toward the laser, increasing absorption to 5.8 mW and the self-shift to 920 pm [Fig. 14(c,d)]. Heater and optical heating therefore act together during acquisition.

For the Z-shaped ring, applying the final 6.9 mW command directly selects a stable branch 684 pm blue of the laser. A temporary 16 mW command moves the resonance through the turning point. Returning to 6.9 mW after that transition reaches the target 142 pm red of the laser. The startup pulse thus selects the operating branch. The hot spectra in Fig. 14 are fixed-temperature probe sweeps with bias-dependent loss, with loss and coupling otherwise temperature independent.

C. Heater tuning

Joule heating raises the waveguide temperature and effective index, shifting the resonance red. For constant heater tuning efficiency,

$$\Delta\lambda_{\text{res}} = \left. \frac{\Delta\lambda_{\text{res}}}{P_h} \right|_{\text{ref}} P_h. \quad (31)$$

where $P_h = V_h^2/R_h$ is heater dissipation, V_h its voltage and R_h its resistance. The reference efficiency is fitted relative to zero heater power. The heater efficiencies of the Z-shaped ring and the 256G lateral ring are 73 and 160 pm/mW, with resistances of 500 and 317 Ω , respectively^{12,36}. At fixed loss and coupling, heater tuning shifts the transmission curve in Eq. (12) without changing its shape.

D. Cold resonance placement

Designed resonance spacing can reduce postfabrication tuning requirements⁵⁵, while self-heating changes the operating wavelength and branch acquisition⁵². Cold placement must account for package temperature, DC bias and optical heating while retaining heater correction in both directions. Ring path length and cross-section set the cold wavelength, whereas coupling and doping set the required hot-state linewidth, notch depth, tuning efficiency and electrical response.

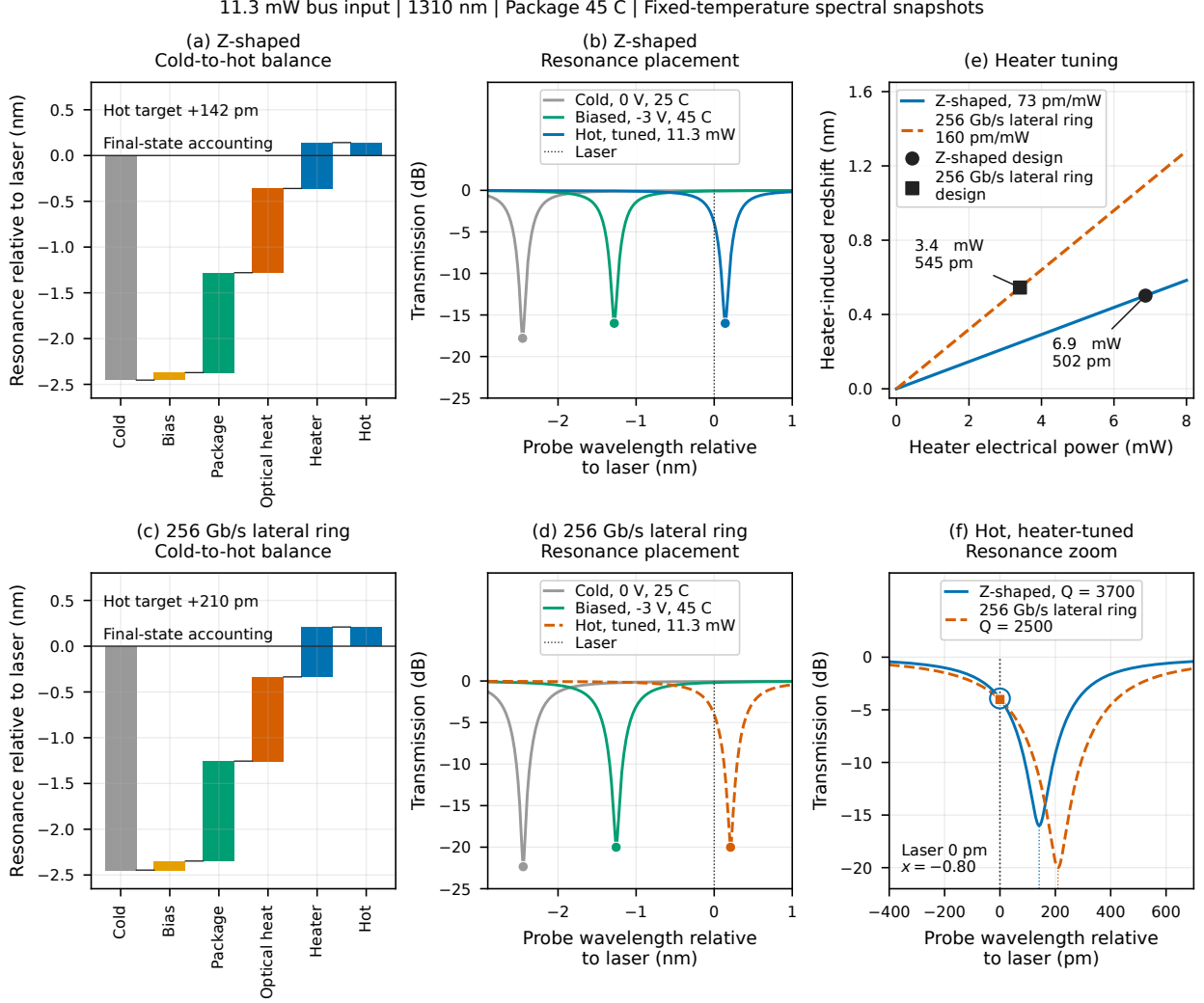


FIG. 14. Conditional resonance placement at 11.3 mW, 1310 nm and 45 °C. **(a,c)** Z-shaped ring/256G lateral ring wavelength balances. **(b,d)** Corresponding frozen-temperature spectra. **(e)** Heater-only shifts from calibrated efficiencies. **(f)** Hot-spectrum zoom at the assumed $x = -0.80$ target. Circles/squares denote Z-shaped ring/256G lateral ring operating points; dotted lines mark the laser.

Relative to the 0 V, 25 °C cold state without optical or heater power, the first-order wavelength contributions add to give

$$\Delta\lambda_{\text{res}} = \underbrace{\lambda_{\text{res},V_{\text{DC}}} - \lambda_{\text{res},V_{\text{ref}}}}_{\text{bias}} + \underbrace{\Delta\lambda_{\text{res}}|_{\Delta T}}_{\text{package}} + \underbrace{\Delta\lambda_{\text{res}}|_{P_h}}_{\text{heater}} + \underbrace{\Delta\lambda_{\text{res}}|_{\bar{P}_{\text{abs}}}}_{\text{self-heating}} + \underbrace{\Delta\lambda_{\text{res}}|_{\text{other}}}_{\text{other}}, \quad (32)$$

where $\lambda_{\text{res}} - \lambda_{\text{laser}} = c_0 + \Delta\lambda_{\text{res}}$, c_0 is the cold offset from the laser, $V_{\text{ref}} = 0$ the cold-reference junction voltage, V_{DC} the signed operating bias, and ΔT the package-temperature rise above 25 °C. Each contribution is evaluated with the other sources fixed. Package, heater and self-heating shifts

follow Eqs. (25), (31) and (26), respectively. The bias-induced shift uses voltage-indexed wavelengths at the same cold thermal reference. The residual $\Delta\lambda_{\text{res}}|_{\text{other}}$ includes additional nonlinear-index and crosstalk shifts, such as those from photocarriers, not already included in the named contributions. The residual is set to zero in the numerical balance below. Modulation-induced chirp follows from the time-dependent output phase of the cavity model, so it need not be added as a separate resonance shift. At the assumed 33 pm/V tuning efficiency, changing from 0 to -3 V gives a $+99$ pm redshift. Package warming from 25 to 45 °C adds $+1092$ pm, while 11.3 mW bus input contributes $+920$ pm of optical self-heating. Balancing these shifts against a designed cold resonance of -2447 pm requires a nominal heater shift of $+545$ pm (3.4 mW) to reach the target hot resonance at $+210$ pm ($x = -0.80$). Separate heater and optical-heating states retain their response times in the cavity model (Sec. V).

In this cold design, self-heating reduces the heater requirement from 9.2 mW without optical heating to 3.4 mW at 11.3 mW bus input, assuming equal optical and heater tuning efficiencies. The modulation excursion is evaluated about this settled hot resonance. In physical mask layout, setting this cold resonance requires adjusting the drawn ring radius by

$$\Delta R = \frac{L_{\text{rt}}}{2\pi} \left(\frac{n_g}{n_{\text{eff}}} \right) \left(\frac{c_0}{\lambda_{\text{res}}} \right), \quad (33)$$

which evaluates to $\Delta R \approx -17$ nm for the 256G lateral ring and -34 nm for the Z-shaped ring. These sub-35 nm offsets fall comfortably within standard immersion-lithography mask grids, exploiting package warming and optical self-heating to eliminate over 12 mW of steady-state electrical heater dissipation per lane. At 256 Gb/s, this 12.6 mW reduction saves 49.1 fJ/bit, lowering active heater overhead from 62.4 to 13.3 fJ/bit and bringing thermal energy dissipation within the same order of magnitude as the electrical driver. For the 200 Gb/s Z-shaped ring, harnessing the same passive thermal assist saves 27.6 mW of equivalent heating, delivering a 137.8 fJ/bit energy-efficiency improvement.

Correction reserve sets the nominal heater command. For the 11.3 mW example, ± 5 K package drift, ± 100 pm fabrication offset and $\pm 10\%$ bus-power variation give a maximum positive shift error of 465 pm. Compensating this error requires 2.9 mW downward adjustment, which, with 0.5 mW reserve, sets the 3.4 mW nominal command. The opposite corner requires 6.3 mW, within the 0–8 mW heater range with 0.5 mW upper reserve. Cold placement therefore sets the steady heater power and correction range, while the startup command determines access to the intended branch.

V. COMPACT MODELING AND EYE-DIAGRAM EVALUATION

A. Physical and Verilog-A models

Commercial spatial solvers characterize optical fields and carrier transport, while circuit tools use compact component models⁵⁶. Compact models couple extracted electrical parasitics to non-linear cavity and thermal dynamics for electronic–photonic co-design⁴⁴. For the segmented Z-shaped and 256G lateral rings, measured or spatially simulated parameter relations connect junction design, coupling and bias to OMA, bandwidth and thermal power in the physical model.

For numerical integration in Python or Verilog-A, Eq. (1) is expressed in the laser-frequency frame. Writing $a(t) = \tilde{a}(t)e^{j\omega_{\text{in}}t}$ and $E_{\text{in,out}}(t) = \tilde{E}_{\text{in,out}}(t)e^{j\omega_{\text{in}}t}$ for a constant reference frequency ω_{in} gives

$$\frac{d\tilde{a}}{dt} = \left[j\Delta\omega - \frac{1}{2\tau_p} \right] \tilde{a} + \frac{\tilde{E}_{\text{in}}}{\sqrt{\tau_e}}, \quad \tilde{E}_{\text{out}} = \tilde{E}_{\text{in}} - \frac{\tilde{a}}{\sqrt{\tau_e}}, \quad (34)$$

where the tildes denote complex envelopes and $\Delta\omega = \omega_r - \omega_{\text{in}}$ is the signed detuning. The envelopes preserve energy, port power, phase and cavity memory without resolving the optical carrier.

Junction resistance, capacitance and the driving circuit determine the voltage across each segment [Fig. 15]. Depletion changes the index and absorption, shifting the resonance and modifying its linewidth and depth through Eqs. (4) and (5). The loss-aware eyes retain voltage-dependent intrinsic loss at fixed external coupling. Temperature-dependent parameters and dynamic photo-carrier effects require additional calibration.

The resonance combines instantaneous voltage tuning with the thermal shifts in Eq. (32). Separate first-order heater and optical-heating states relax toward Eqs. (31) and (26), with absorption determined by stored cavity energy and locally absorbing loss. Thermal relaxation is much slower than a symbol period, so the eye calculations use the equilibrium in Eq. (27) at average absorption. The squared magnitude of the through-port field gives optical power, including interference, from which OMA, ER and sampled eye opening are extracted.

Verilog-A can express the real and imaginary envelope equations using its built-in time-derivative operator `ddt()`. The circuit simulator solves these differential equations together with the junction current and charge relations and the transistor-level EIC and package network. Matching spectra and waveforms under identical conditions would verify the translation. The measurement comparisons here use Python.

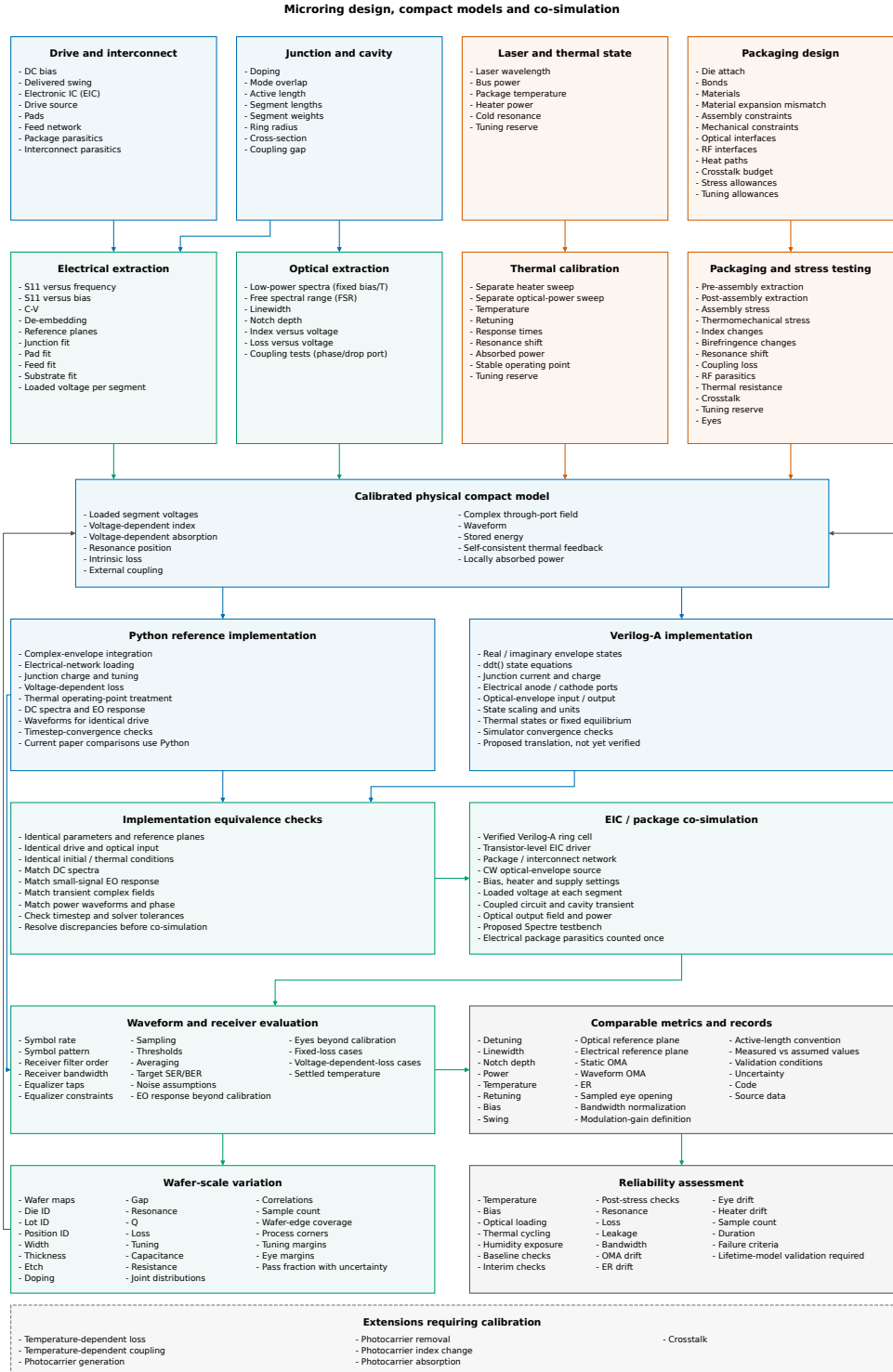


FIG. 15. Microring design, compact models and co-simulation workflow. The dashed box denotes extensions requiring calibration.

B. Device calibration and design tolerance

De-embedded complex S-parameters constrain the electrical network, as in the Z-shaped ring model [Fig. 6] and bias-dependent circuit extraction^{12,44}. For real reference impedance Z_0 , the input impedance follows $Z_{\text{in}} = Z_0(1 + S_{11})/(1 - S_{11})$. Accurate circuit-level co-design requires de-embedding reference planes to the junction boundary to decouple pad capacitance C_{pad} and feed inductance L_p from the active junction impedance R_s and $C_j(V)$. Un-deembedded one-port measurements lump packaging and pad shunts into the junction, masking high-frequency inductive peaking and distorting the extracted capacitance. Multi-length test structures, such as standalone test diodes and ring segments with varying active arc lengths, resolve correlated parameters between junction sheet resistance and metal contact resistance, recovering the true junction-voltage transfer function $H_e(\omega_m)$ in Eq. (22).

Complementary optical parameter extraction must be performed in the linear, weak-probe regime (input power below -10 dBm) to prevent optical self-heating from distorting the extracted linewidth and notch depth. Group index n_g , loaded quality factor Q_L , and notch depth are determined from through-port transmission via Eqs. (8), (3) and (5). Because through-port intensity transmission exhibits identical Lorentzian dips for undercoupled and overcoupled cavities, designers must resolve the coupling regime unambiguously by measuring transmission phase, probing an auxiliary drop port, or tracking the notch depth derivative under reverse bias. Voltage sweeps then constrain the electro-optic phase efficiency $\eta(V)$ and voltage-dependent absorption $\Delta\alpha(V)$, while thermal steps constrain heater tuning efficiencies and thermal response times for large-signal Verilog-A tables.

Translating device-level calibration to multi-lane DWDM arrays requires managing dimensional fabrication tolerances. In silicon photonics foundry processes, across-wafer waveguide width and thickness exhibit typical standard deviations of $\sigma_w \approx 4.6$ nm and $\sigma_h \approx 0.8$ nm, with systematic wafer-scale gradients accounting for roughly 80% of the total variance⁵⁷. Because each nanometer of width deviation shifts the resonance by approximately 1 nm, uncompensated fabrication skew can exceed the entire channel spacing of a dense DWDM grid. Array layouts must therefore place microring modulators in immediate spatial proximity with matched dummy pattern fills and identical orientation. This tight layout matching suppresses intra-array relative skew to below 100 pm, confining active thermal tuning to common-mode tracking rather than wide per-channel correction.

Passive tolerance engineering can substantially desensitize the cavity to etching and lithographic variations before active compensation. Incorporating adiabatically widened waveguide bends in non-modulating arcs reduces optical mode interaction with etched sidewall roughness, cutting across-wafer resonance wavelength standard deviation by a factor of 2.1 relative to uniform narrow rings⁵⁸. In the bus coupling region, shallow-etched ridge couplers provide up to a fourfold reduction in the normalized standard deviation of the coupling coefficient compared to deep-etched strip couplers⁵⁹, stabilizing the power transfer fraction κ^2 against over- or undercoupling drift. For active modulators, width transitions between widened passive arcs and narrow active junctions must remain strictly adiabatic to preserve fundamental mode purity and avoid modal scattering loss.

Statistical yield must be evaluated at the link level rather than through isolated component tolerances⁶⁰. In dense co-packaged transmitters, the available on-chip heater power (typically 0 to 8 mW per lane) sets a strict upper bound on correctable resonance error. Monte Carlo co-simulation should jointly sample the correlated distributions of junction capacitance, series resistance, cold resonance offset and tuning efficiency against receiver sensitivity and link loss. To prevent thermal runaway or excessive interchannel thermal crosstalk, transceivers should implement algorithmic wavelength arbitration, such as cyclic locking or nearest-line assignment, which remaps logical data streams to adjacent optical carrier lines and cuts worst-case thermal tuning power by more than 60% compared to fixed-grid locking.

Design-for-test architectures must separate fabrication variations from packaging-induced assembly shifts. Dedicated grating couplers enable automated wafer-level optical probing to screen cold resonance wavelengths, loaded quality factors and DC extinction ratios before committing dies to 2.5D or 3D integration⁷. Subsequent assembly introduces parasitic shifts, including 20 to 50 pH of micro-bump or wirebond inductance and thermal dissipation from adjacent CMOS driver dies. Characterizing test structures before and after packaging isolates package-induced electrical loading from intrinsic silicon process drift, confirming that the high-speed eye margin remains intact.

During live operation, closed-loop feedback stabilizes the modulator against ambient temperature drift⁶¹. Control loops monitoring through-port average power or dedicated tap photodiodes dynamically trim micro-heaters to lock the operating detuning near the target optimum ($x \approx -0.80$). The feedback algorithm must account for the asymmetric thermal stability between the self-stabilizing blue-detuned flank and the regenerative red-detuned flank to prevent lock run-

away. Over operating lifetimes, high intra-cavity optical intensity and sustained reverse-bias electric fields can induce charge trapping in the oxide or slab interfaces, causing slow threshold shifts. Transceiver co-design should allocate a 10% to 15% overdrive margin in the CMOS driver swing to absorb long-term efficiency degradation without violating optical modulation amplitude specifications.

C. Validation and eye-quality analysis

1. Measured and modeled responses

For our Z-shaped ring, the model with its electrical network extracted from segment S_{11} [Fig. 6] gives 49 GHz EO bandwidth, matching the measured value¹². The 200 Gb/s eyes in Fig. 16 compare measured and modeled responses.

The measured PAM4 eyes use two 100 Gb/s NRZ drives at 1.6 V_{pp} per segment after the bias-tees. The measured outer ER is 3.6 dB and TDECQ is 0.2 dB at an SER threshold of 10^{-2} with a 21-tap FFE, applying Bessel filtering and 64-fold pattern averaging to suppress amplified spontaneous emission (ASE)¹².

The simulations use equal 1.6 V_{pp} drives and an assumed 9/18 pm/V tuning split motivated by approximately two-to-one geometry and capacitance. Panels (e–g) use single acquisitions with noise and 21-tap FFE, without pattern averaging. One amplitude calibration to the measured LSB rail separation is shared across simulations at 230 μ W and 5 ps per division. Differences in processing limit comparisons of absolute rail spacing.

2. Detuning and eye quality

Increasing detuning broadens the optical response at the cost of low-frequency gain and greater overshoot. The small-signal response is shown in Fig. 9, where cavity peaking grows with detuning while the loaded electrical response of Eq. (22) limits the cascade, so the modeled electro-optic bandwidth shows no net peaking. Figure 17 connects this detuning dependence to large-signal metrics and time-domain distortion across laser detuning x . Static extinction ratio [Fig. 17(a)] increases as the laser approaches the resonance notch. Because reverse bias widens depletion and red-shifts the resonance ($\Delta x > 0$), it drives the lower rail directly into the notch on the red side ($x > 0$), producing a steeper static extinction ratio than on the blue side. Dynamic optical mod-

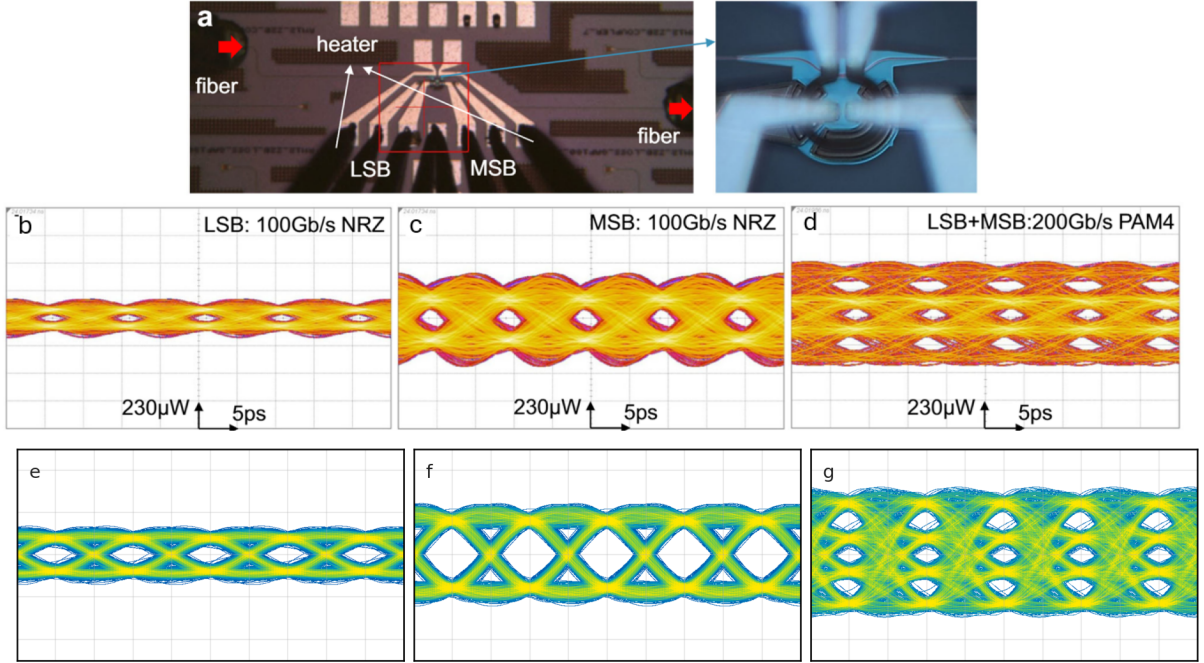


FIG. 16. Z-shaped ring transmission. **(a)** Device illustration and micrograph. **(b,c)** Measured 100 Gb/s LSB/MSB NRZ eyes. **(d)** Measured 200 Gb/s PAM4. **(e–g)** Simulated LSB, MSB and PAM4 eyes with noise and 21-tap FFE, without measurement averaging. Panels (a)–(d) cropped/rescaled from Yuan *et al.*, Fig. 4¹², CC BY 4.0.

ulation amplitude under voltage-dependent loss [Fig. 17(b)] instead favors the blue side ($x < 0$), peaking near $x \approx -0.58$ for the Z-shaped ring and $x \approx -0.80$ for the 256G lateral ring. Here, resonance red-shift and reduced free-carrier absorption combine constructively to increase transmission, whereas on the red side competing effects reduce dynamic OMA by 5% to 8%. Figure 17(c) carries this trade-off into the link domain, comparing required optical bus power $P_{\text{bus,req}}$ on the left axis against limiting step overshoot on the right axis. Junction parasitics shift these operating points relative to the bare optical cavity. At DC, vanishing capacitive displacement current leaves the maximum-slope point locked at the stationary Lorentzian inflection point ($x = -0.58$). Under 128 GBaud modulation, the dynamic OMA peak shifts outward to $x \approx -0.80$ as optical cavity peaking compensates for junction RC roll-off, while electrical filtering pulls the maximum-GBW bias inward from the bare-cavity value ($|x| = 1.13$) to $x = -0.98$.

Step overshoot reaches 24% for the 256G lateral ring and 27% for the Z-shaped ring at $x = -0.80$, increasing to 36% at $x = -0.98$. Required bus power forms a bathtub minimum near

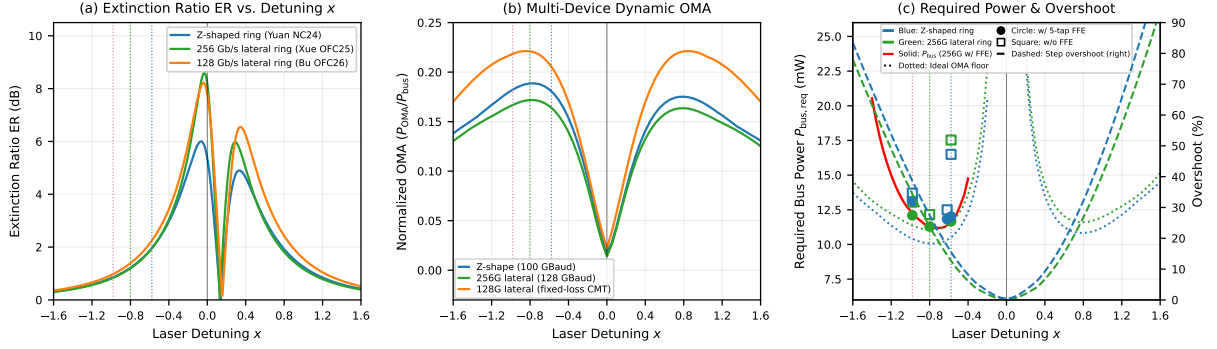


FIG. 17. Detuning trade-offs across extinction ratio, optical modulation amplitude, and link power. **(a)** Static two-rail extinction ratio ER. **(b)** Large-signal dynamic OMA across three reference devices. **(c)** Required bus power $P_{bus, req}$ and limiting step overshoot under voltage-dependent loss. Dotted vertical lines mark maximum slope ($x = -0.58$, blue), maximum dynamic OMA ($x = -0.80$, green), and circuit-loaded maximum GBW ($x = -0.98$, pink).

$x \approx -0.80$, reaching 11.3 mW for the 256G lateral ring and 11.8 mW for the Z-shaped ring. Dotted curves trace the ideal-OMA floor without intersymbol interference. Without receiver equalization, intersymbol interference increases the power required at maximum slope to 17.5 mW (squares), whereas a five-tap equalizer lowers required power to 11.7 mW (circles), providing a 5.8 mW equalization recovery.

The detuning trade-off also affects level spacing and noise tolerance. The noise comparisons use the fixed-loss 256G lateral-ring model (Sec. III D), with a refined GBW bias at $x = -0.98$. Table IV gives the sweep at 11.3 mW bus input. Peaking and rise metrics use small-signal and step responses, while OMA and ER use waveform extrema. Despite greater overshoot at the bare-cavity GBW bias, its five-tap equalized penalty is close to that at maximum slope. Fitted level linearities are 97% at the bare-cavity GBW bias and 89% at maximum slope. Rise times are percentages of the unit interval (UI), the symbol duration.

Across the held-detuning sweep, waveform OMA varies by 7% at 1.8 V_{pp} and 5% at 4 V_{pp}. Detuning directly affects link power margin and level spacing, requiring thermal stabilization to preserve the target operating point. The fixed-loss sweeps exclude capture dynamics, power-dependent loss and carrier distortion.

TABLE IV. Fixed-loss eye properties at 128 GBaud and 1.8 V_{pp}, normalized to 11.3 mW bus input. Penalties are not compliance TDECQ.

Operating point	$ x $	Peaking	Overshoot	Rise	ER	OMA	TDECQ
		(dB)	(%)	(% UI)	(dB)	(% P_{in})	(10 ⁻² dB)
Maximum small-signal slope	0.58	0.3	13	38	2.8	16	8
Reported bias	0.98	2.5	36	19	1.5	16	11
Near bare-cavity GBW optimum	1.13	3.5	47	16	1.2	16	14
Beyond the optimum	1.50	6.2	76	9	0.9	14	18

3. Noise loading and TDECQ

TDECQ expresses the eye distortions above as a noise-tolerance penalty, comparing ideal and device noise tolerances at the same outer OMA. The simplified estimator is $10\log_{10}(\sigma_{ideal}/\sigma_{eq})$. Here $\sigma_{ideal} = OMA_{outer}/(6Q_t)$ and σ_{eq} is the Gaussian noise RMS at the target SER. A fourth-order half-baud Bessel–Thomson receiver precedes delay compensation and equalization, with equalized level means setting thresholds and outer OMA. This receiver and its five-tap symbol-spaced equalizer follow the reference-receiver and reference-equalizer convention of the IEEE 802.3 transmitter test⁶². The fixed reference receiver isolates transmitter quality and leaves the penalty independent of the photodiode a link later uses. Compliance TDECQ instead requires prescribed outer-level extraction, receiver, equalizer constraints and impairment treatment.

For the fixed-loss 256G lateral ring, $Q_t = 3.4$ and target SER 4.8×10^{-4} give five-tap penalties of 0.1 dB at 1.8 V_{pp} and 0.6 dB at 4 V_{pp}. Twenty-one taps change both by 0.01 dB.

Reported TDECQ values of 1.6 and 3.9 dB at 224 and 240 Gb/s PAM4³⁴ were taken at different symbol rates of 112 and 120 GBaud, so the higher baud rate accounts for most of the gap, with receiver filtering a secondary effect. Because TDECQ depends on the reference receiver and equalizer prescribed by each interface standard, the two numbers are compliance results under different specifications and are not directly comparable.

Receiver filtering can change the reported penalty for an unchanged optical waveform. At 128 GBaud, 1.8 V_{pp} and the reference bias $x = -0.98$, the fixed-loss 256G lateral ring gives simplified penalties of 1.7 and 0.1 dB with fourth-order Bessel receivers at approximately 27 and

64 GHz. Both use five unconstrained symbol-spaced least-squares taps and the same phase and threshold algorithms, refitted after filtering with the same record for fitting and evaluation. Increasing resolution from 32 to 64 samples per symbol changes either penalty by less than 0.1 dB. An identical-waveform comparison with approximately 40 and 96 GHz receivers gives about 0.2 dB in both cases, illustrating the dependence on the waveform and evaluation procedure.

Figure 18 compares the 256G lateral-ring model at 128 GBaud and 1.8 V_{pp} with the Z-shaped ring model at 100 and 128 GBaud and 1.6 V_{pp} per segment. The latter uses $Q_L = 3700$, fitted segment circuits with a 50 Ω source, an assumed 9/18 pm/V tuning split and PRBS9 streams, versus random PAM4 for the 256G lateral ring. Both use fixed intrinsic loss, half-baud receivers and five-tap equalizers. Panels (a,b) use the estimator above at the common reference bias $x = -0.98$ for all three device/rate cases. Panel (a) adds multiplicative output RIN, while the drive sweep in panel (b) excludes RIN. Panels (c,d) propagate input RIN and subtract the ideal-reference penalty for each device and rate. Panel (c) retains all three 256G lateral-ring biases and only the maximum-OMA bias for the Z-shaped ring at each rate. Panel (d) includes all three biases for each case. The drive sweep extrapolates the fixed-loss model.

Sequentially adding the electrical, cavity and RIN responses to rectangular PAM4 quantifies their incremental noise penalties [Fig. 18(a)]. At 128 GBaud, the 256G lateral ring's internal electrical filtering contributes less than 0.01 dB, versus 0.1 dB from the coupled cavity memory and flank curvature and 0.5 dB from output RIN. For the Z-shaped ring at 100 GBaud, the corresponding cavity and RIN increments are 0.1 and 0.4 dB. Their 0.5 dB sum excludes the 0.024 dB electrical-only increment. At 128 GBaud, the Z-shaped ring's cavity and output-RIN increments increase to 0.2 and 0.8 dB, respectively, again excluding the electrical-only increment. RIN dominates the cavity contribution under these assumptions.

At the nominal drives above, moving from $x = -0.98$ to the maximum-dynamic-OMA bias gives a larger optical modulation amplitude with a smaller output-RIN penalty. The 256G lateral ring at 128 GBaud and the Z-shaped ring at 100 and 128 GBaud have maximum-OMA biases of $x = -0.80$, -0.62 and -0.63 , respectively. Their waveform-extrema OMA increases by 2.9%, 12.0% and 14.4%, while the penalties without RIN decrease by 0.042, 0.015 and 0.011 dB with the same estimator. At -155 dB/Hz output RIN, the summed cavity and RIN penalties decrease from 0.6 to 0.3 dB, 0.5 to 0.2 dB and 1.0 to 0.5 dB, respectively, retaining the electrical-only subtraction of panel (a). The lower mean transmitted power at maximum OMA reduces multiplicative output noise at fixed bus input power.

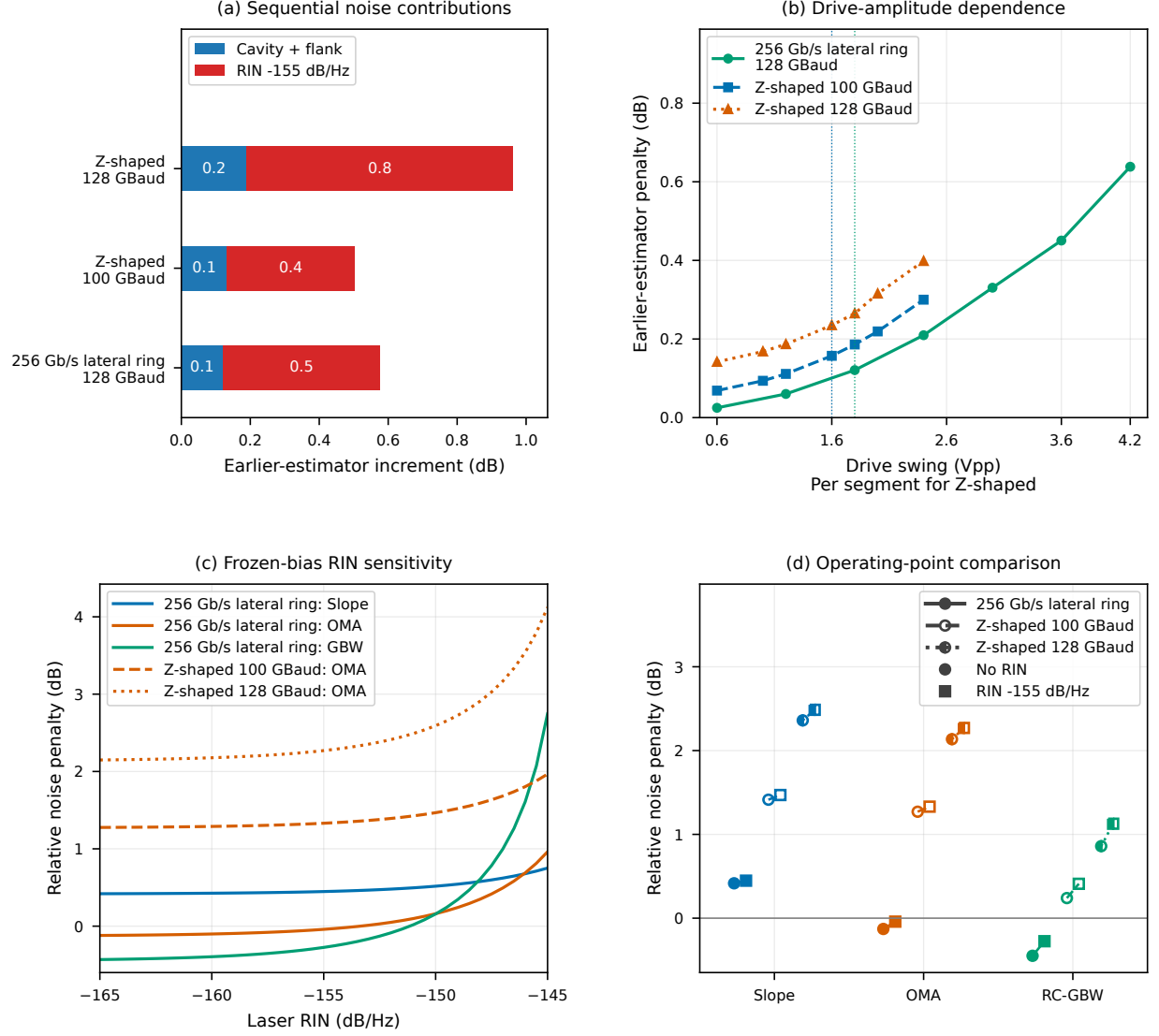


FIG. 18. Modeled noise penalties, not compliance TDECQ. **(a,b)** Cavity/RIN contributions and drive sweep at $|x| = 0.98$, near the maximum GBW bias, for all three device/rate cases. **(c)** RIN sensitivity at three 256G lateral-ring biases and maximum OMA for the Z-shaped ring at 100/128 GBaud. **(d)** All three biases for each case, without and with -155 dB/Hz RIN. Panels (c,d) subtract each case's ideal reference; the Z-shaped ring results at 128 GBaud are model extrapolations.

The Z-shaped ring has higher equalized PAM4 level uniformity at $x = -0.98$. This uniformity, the level-separation-mismatch ratio R_{LM} fitted from the four sampled power-level means after receiver filtering and equalization, changes from 96.4% to 92.5% at 100 GBaud and from 97.8% to 92.2% at 128 GBaud when moved to maximum OMA. A broader response can reduce intersymbol

interference, while detuning also changes the voltage-to-power curvature sampled by the four levels. The fitted level spacing reflects both effects together with receiver filtering and equalization. The common reference bias therefore provides higher equalized PAM4 level uniformity, while maximum OMA provides greater optical modulation amplitude and a lower output-RIN penalty under these conditions.

At the 256G lateral ring’s fixed drive-sweep bias, optical bandwidth exceeds twice Nyquist, leaving little residual ISI for additional taps to remove. Increasing swing instead samples more Lorentzian curvature and compresses outer levels. The equalized PAM4 level uniformity R_{LM} falls from 99% to 96%, approaching the 95% comparison criterion. The smaller drive reduces the penalty but gives less than half the OMA, limiting tolerable link loss. Drive and bias must therefore meet OMA, level-spacing and noise requirements together.

4. *Relative intensity noise*

Unlike the output-RIN loading above, the input-RIN comparison includes cavity filtering. Figure 18(c,d) compares three fixed-loss operating points of the 256G lateral ring with a 64 GHz receiver and five-tap symbol-spaced equalizer. Taps and thresholds fitted on the training half are frozen during evaluation, with equalizer noise enhancement included. Subtracting the 0.4 dB filtered-ideal-PAM4 penalty gives signed values, with negative values indicating improvement over that reference under unconstrained equalization.

White input RIN is propagated through the frozen-bias cavity, receiver and equalizer over 0–64 GHz. At -155 dB/Hz, the signed penalties change from 0.42 to 0.45 dB at maximum slope, from -0.13 to -0.041 dB at maximum OMA, and from -0.45 to -0.28 dB at maximum GBW [Fig. 18(d)]. GBW’s larger increment is consistent with higher mean transmission and lower contrast. The calculation excludes noise above 64 GHz, aliasing and level-dependent conversion during modulation.

The Z-shaped ring comparison at 100 GBaud uses the undercoupled branch and both fitted segment networks with a $50\ \Omega$ source, the same training/evaluation split, a 50 GHz Bessel receiver and five-tap equalizer. Its own 0.4 dB ideal-reference penalty is subtracted. At slope, dynamic-OMA and circuit-loaded GBW biases of $|x| = 0.58, 0.62$ and 0.97 , the baseline penalties are 1.4, 1.3 and 0.2 dB. Input RIN at -155 dB/Hz adds 0.056, 0.061 and 0.17 dB, respectively, over 0–50 GHz under the frozen-bias approximation. The normalized comparison favors circuit-loaded

GBW at the demonstrated 100 GBaud rate, while link-power selection also requires absolute OMA and sampled level separation.

At 128 GBaud, the same Z-shaped ring model uses a 64 GHz receiver with input RIN integrated over 0–64 GHz. The slope, rate-specific waveform-OMA and circuit-loaded GBW biases are $|x| = 0.58, 0.63$ and 0.97 , with relative penalties of 2.4, 2.1 and 0.9 dB without RIN, rising to 2.5, 2.3 and 1.1 dB at -155 dB/Hz. The five-tap training/evaluation procedure is unchanged. Increasing resolution from 64 to 128 samples per symbol changes these penalties by less than 0.02 dB. The higher-rate case is a fixed-loss model extrapolation, not the demonstrated 200 Gb/s (100 GBaud) Z-shaped ring experiment.

Absolute noise tolerance also depends on level separation. For the 256G lateral ring under the common calibration in Sec. III D 2, maximum OMA requires the least power despite GBW’s lower normalized penalty, as shown by the required-power comparison at 128 GBaud in that section. Minimum equalized eyes are 90%, 86% and 88% of the ideal adjacent-level gap at OMA, slope and GBW, respectively. Absolute level separation, residual ISI and RIN jointly determine sensitivity through Eq. (29).

Transmitter pre-emphasis is also constrained by the available swing. For the 256G lateral ring, peak-limited two-tap TX searches at 128 GBaud select bypass for OMA and GBW with five-tap RX FFE. At the $1.8 V_{pp}$ command limit, precursor/postcursor emphasis reduces sustained drive during long symbol runs. Figure 13 retains bypass, with slope and higher-rate TX settings unoptimized.

VI. CONCLUSION AND OUTLOOK

Interconnect scaling is leading the performance and energy footprint of distributed artificial intelligence fabrics. Depletion-mode silicon microrings have proven their capability by delivering 200 Gb/s PAM4 with record-low switching energy, yet scaling to 400G per lane and multi-Terabit escape densities demands a fundamental departure from isolated component design toward integrated electronic–photonic co-design. Optical modulators and electronic transceivers must therefore be designed together as a single system. The physical and Verilog-A models developed in this review provide the unified framework to enable this co-design.

Scaling individual lanes toward 400G at 212.5 GBaud PAM4 requires balancing photon lifetime against resonant enhancement through controlled overcoupling and detuning-peaking. High-

bandwidth avalanche photodiodes provide a complementary route to overcome electronic receiver noise under high-speed operating photocurrents. Across multi-channel transmitter arrays, dense wavelength-division multiplexing must scale concurrently through automated thermal locking and channel allocation to suppress interchannel crosstalk as aggregate bandwidth density increases.

Beyond device and wavelength scaling, high-speed optical links face an emerging gap between interoperability standards and real-world deployment needs. Specifications such as IEEE 802.3dj define baseline compliance thresholds, yet hyperscale datacenters often require pre-FEC bit-error rates near 10^{-9} to maintain mission-critical application performance. Quantifying and protecting this unallocated implementation margin represents an essential co-design problem for the industry.

Managing that margin over dynamic operating conditions points toward adaptive, closed-loop optical interfaces. Frameworks such as autonomous path startup (APSU) and inter-sublayer link training (ILT) can be extended from the electrical host interface into the optical domain. Using far-end receiver telemetry to dynamically adjust transmitter operating points creates an actionable route toward continuous link-margin tracking and in-service retraining without interrupting live data traffic.

The predictive modeling framework presented here bridges device physics with circuit and link design. By unifying physical device equations, Verilog-A circuit co-simulation, and link-budget evaluation, this framework equips Tier-1 network system architects to balance power, thermal, and reach trade-offs across distributed AI fabrics, while enabling photonic and electronic designers to systematically explore the multidimensional co-design space to scale optical interconnects to 400G per lane and multi-Terabit capacities.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

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